CSE thesis random forest P3

January 24, 2026

[1]:
```
"""
Reusable Random Forest Pipeline for Microfinance Impact Analysis
----------------------------------------------------------------
Use this to predict BF1 / BF3 / BF4 / BF5 ... by changing ONE variable.

Example:
    run_comprehensive_bf_analysis(target_bf="BF1")
    run_comprehensive_bf_analysis(target_bf="BF3")
"""

# ===========
# 0) IMPORTS
# ===========
import os, io, re, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪cross_validate
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    accuracy_score, precision_score, recall_score, f1_score
)
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
import joblib

RANDOM_STATE = 42
```

```python
# ============================================================
# 1) DATA LOADING
# ============================================================
def load_excel_from_colab():
    """Upload Excel in Google Colab."""
    try:
        from google.colab import files
        print("  Please select the Excel file with your encoded dataset...")
        uploaded = files.upload()
        path = list(uploaded.keys())[0]
        df = pd.read_excel(path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        raise RuntimeError("Colab upload failed.") from e


def load_excel_locally():
    """Upload Excel when running locally."""
    try:
        import tkinter as tk
        from tkinter import filedialog
        print("  Please select the Excel file...")
        root = tk.Tk()
        root.withdraw()
        file_path = filedialog.askopenfilename(
            title="Select Excel file",
            filetypes=[("Excel files", "*.xlsx *.xls")]
        )
        if not file_path:
            raise ValueError("No file selected")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        print(f"  File dialog failed: {e}")
        file_path = input("  Enter full path to your Excel file: ")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df


# ============================================================
# 2) TARGET SELECTION (GENERALIZED BFx)
# ============================================================
def detect_bf_target(df: pd.DataFrame, target_bf: str = "BF1") -> str:
    """
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```
    Detect BF target column robustly.
    target_bf examples: "BF1", "BF3", "BF5"
    """
    m = re.match(r"^\s*BF\s*(\d+)\s*$", str(target_bf), flags=re.I)
    if not m:
        raise ValueError(f"target_bf must look like 'BF1', 'BF3', etc. Got:␣
↪{target_bf}")

    k = m.group(1)  # number as string
    aliases = [
        f"BF{k}", f"bf{k}",
        f"BF {k}", f"bf {k}",
        f"BF_{k}", f"bf_{k}",
        f"BF-{k}", f"bf-{k}",
    ]

    for a in aliases:
        if a in df.columns:
            print(f"  Found target column: {a}")
            return a

    # case-insensitive regex search in column names
    pat = re.compile(rf"\bbf\s*[_-]?\s*{k}\b", flags=re.I)
    for col in df.columns:
        if pat.search(str(col)):
            print(f"  Found target column (regex match): {col}")
            return col

    print(f"  {target_bf} column not found automatically.")
    print("  Available columns:")
    print(df.columns.tolist())
    user_col = input(f"  Enter exact column name for {target_bf}: ").strip()
    if user_col in df.columns:
        print(f"  Using specified target column: {user_col}")
        return user_col
    raise ValueError(f"Column '{user_col}' not found in dataset")


def preprocess_bf_target(y_series: pd.Series, target_label: str):
    """
    Convert BF target responses into binary classification (expects 0/1 after␣
↪cleaning).
    """
    y_numeric = pd.to_numeric(y_series, errors="coerce")
    valid_mask = y_numeric.notna() & np.isfinite(y_numeric)
    y_clean = y_numeric[valid_mask]
```

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    if len(y_clean) == 0:
        raise ValueError(f"  No valid values for {target_label} after cleaning")

    print(f"\n  TARGET ANALYSIS: {target_label}")
    print("-" * 40)
    print("Value distribution:")
    print(y_clean.value_counts().sort_index())

    unique_vals = sorted(y_clean.unique())
    print(f"Unique values: {unique_vals}")

    # If your encoded dataset is already 0/1, this is perfect.
    # If it includes 1/2 or 1/3 scales, you should define a threshold rule here.
    y_binary = y_clean.astype(int)

    class_balance = y_binary.value_counts(normalize=True).round(3)
    print(f"  Final class balance: {class_balance.to_dict()}")

    return y_binary, valid_mask


# ============================================================
# 3) FEATURE ENGINEERING
# ============================================================
def build_feature_matrix(df_clean: pd.DataFrame, target_col: str):
    """Prepare features for ML (numeric-only)."""
    print("\n  FEATURE ENGINEERING")
    print("-" * 40)

    X = df_clean.drop(columns=[target_col], errors="ignore")
    numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
    print(f"  Found {len(numeric_cols)} numeric features")

    if len(numeric_cols) == 0:
        raise ValueError("  No numeric features found for modeling")

    print("  Handling missing values with median imputation...")
    imputer = SimpleImputer(strategy="median")
    X_imputed = imputer.fit_transform(X[numeric_cols])
    X_imputed = pd.DataFrame(X_imputed, columns=numeric_cols, index=X.index)

    non_constant_cols = [c for c in X_imputed.columns if X_imputed[c].nunique()␣
 ↪> 1]
    dropped = len(numeric_cols) - len(non_constant_cols)
    if dropped > 0:
        print(f"   Removed {dropped} constant columns")
```

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    X_final = X_imputed[non_constant_cols]
    print(f"  Final feature matrix: {X_final.shape} (rows, cols)")
    return X_final, non_constant_cols


# ============================================================
# 4) CROSS-VALIDATION
# ============================================================
def perform_cross_validation(X, y, cv_folds=10):
    print(f"\n  PERFORMING {cv_folds}-FOLD CROSS-VALIDATION")
    print("=" * 50)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    scoring = {
        "accuracy": "accuracy",
        "precision": "precision",
        "recall": "recall",
        "f1": "f1",
        "roc_auc": "roc_auc",
        "average_precision": "average_precision"
    }

    cv_results = cross_validate(
        pipe, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,␣
 ↪random_state=RANDOM_STATE),
        scoring=scoring,
        return_train_score=True,
        n_jobs=-1
    )
```

```
    print("  CROSS-VALIDATION RESULTS (Mean ± Std):")
    print("-" * 40)

    metrics_summary = {}
    for metric in scoring.keys():
        test_scores = cv_results[f"test_{metric}"]
        train_scores = cv_results[f"train_{metric}"]

        test_mean = float(np.mean(test_scores))
        test_std = float(np.std(test_scores))
        train_mean = float(np.mean(train_scores))

        metrics_summary[metric] = {
            "test_mean": test_mean,
            "test_std": test_std,
            "train_mean": train_mean
        }

        print(f"{metric.upper():<15}: {test_mean:.4f} ± {test_std:.4f} (train:␣
↪{train_mean:.4f})")

    overfit_gap = metrics_summary["accuracy"]["train_mean"] -␣
↪metrics_summary["accuracy"]["test_mean"]
    print(f"\n  Overfitting Analysis:")
    print(f"   Train-Test Accuracy Gap: {overfit_gap:.4f}")
    print("     Potential overfitting detected" if overfit_gap > 0.05 else "  ␣
↪  Model generalizes well")

    return metrics_summary, cv_results


# ============================================================
# 5) TRAIN/TEST/VALIDATION SPLITS + TRAINING
# ============================================================
def create_validation_set(X, y, validation_size=0.2):
    print(f"\n  CREATING VALIDATION SET ({validation_size*100:.0f}% of data)")
    print("=" * 50)

    X_train_test, X_val, y_train_test, y_val = train_test_split(
        X, y, test_size=validation_size, stratify=y, random_state=RANDOM_STATE
    )

    X_train, X_test, y_train, y_test = train_test_split(
        X_train_test, y_train_test, test_size=0.25, stratify=y_train_test,␣
↪random_state=RANDOM_STATE
    )
```

```
    print(f"  Training set: {X_train.shape[0]} samples")
    print(f"  Test set: {X_test.shape[0]} samples")
    print(f"  Validation set: {X_val.shape[0]} samples")

    return X_train, X_test, X_val, y_train, y_test, y_val


def train_with_comprehensive_validation(X, y, use_validation_set=True):
    print("\n  MODEL TRAINING: Random Forest")
    print("-" * 40)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    if use_validation_set:
        X_train, X_test, X_val, y_train, y_test, y_val =␣
 ↪create_validation_set(X, y)
        print("  Training model with regularization...")
        pipe.fit(X_train, y_train)
        return pipe, X_train, X_test, X_val, y_train, y_test, y_val

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, stratify=y, random_state=RANDOM_STATE
    )
    print(f"  Training set: {X_train.shape[0]} samples")
    print(f"  Test set: {X_test.shape[0]} samples")
    print("  Training model with regularization...")
    pipe.fit(X_train, y_train)
    return pipe, X_train, X_test, None, y_train, y_test, None
```

```python
# ============================================================
# 6) EVALUATION
# ============================================================
def evaluate_model_comprehensive(model, X_test, y_test, dataset_name="Test"):
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    print("\n" + "=" * 60)
    print(f"  {dataset_name.upper()} SET COMPREHENSIVE EVALUATION")
    print("=" * 60)
    print(f"  Accuracy:  {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall:    {rec:.4f}")
    print(f"  F1-Score:  {f1:.4f}")
    print(f"  ROC-AUC:   {roc_auc:.4f}")
    print(f"  PR-AUC:    {pr_auc:.4f}")

    print(f"\n {dataset_name} Classification Report:")
    print(classification_report(y_test, y_pred, digits=3))

    cm = confusion_matrix(y_test, y_pred)
    print(f"  {dataset_name} Confusion Matrix:")
    print("[[TN FP]\n [FN TP]]")
    print(cm)

    return y_pred, y_proba, {
        "accuracy": acc, "precision": prec, "recall": rec, "f1": f1,
        "roc_auc": roc_auc, "pr_auc": pr_auc
    }


# ============================================================
# 7) VISUALIZATIONS (UPDATED COLORS ADDED)
# ============================================================
def create_validation_comparison_plot(cv_metrics, output_dir, prefix):
    metrics_names = list(cv_metrics.keys())
    test_means = [cv_metrics[m]["test_mean"] for m in metrics_names]
    test_stds = [cv_metrics[m]["test_std"] for m in metrics_names]

    plt.figure(figsize=(12, 6))
```

```
    x_pos = np.arange(len(metrics_names))

    # ---- UPDATED: bars have different colors automatically ----
    bars = plt.bar(
        x_pos,
        test_means,
        yerr=test_stds,
        capsize=5,
        alpha=0.7
    )
    for i, b in enumerate(bars):
        b.set_color(f"C{i}")
    # ------------------------------------------------------------

    plt.xlabel("Performance Metrics")
    plt.ylabel("Score")
    plt.title(f"{len(metrics_names)} Metrics Cross-Validation Performance␣
 ↪({prefix.upper()})")
    plt.xticks(x_pos, [m.upper() for m in metrics_names], rotation=0)
    plt.ylim(0, 1)
    plt.grid(True, alpha=0.3, axis="y")

    for i, (mean, std) in enumerate(zip(test_means, test_stds)):
        plt.text(i, mean + 0.02, f"{mean:.3f}±{std:.3f}", ha="center",␣
 ↪va="bottom", fontsize=10)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_cross_validation_results.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()


def create_comprehensive_visualizations(y_test, y_proba, model, feature_names,␣
 ↪output_dir, prefix, cv_metrics=None):
    os.makedirs(output_dir, exist_ok=True)

    rf_model = model.named_steps["rf"]
    imp_df = pd.DataFrame({
        "feature": feature_names,
        "importance": rf_model.feature_importances_
    }).sort_values("importance", ascending=False)

    imp_df.to_csv(f"{output_dir}/{prefix}_feature_importances.csv", index=False)

    # Feature importance plot
    plt.figure(figsize=(12, 8))
    top_features = imp_df.head(15).iloc[::-1]
```

```
    plt.barh(range(len(top_features)), top_features["importance"], alpha=0.7)
    plt.yticks(range(len(top_features)), top_features["feature"])
    plt.xlabel("Feature Importance Score")
    plt.title(f"Top 15 Features for {prefix.upper()} Prediction")
    plt.grid(True, alpha=0.3, axis="x")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_feature_importances.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # ROC + PR curves
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    roc_auc = roc_auc_score(y_test, y_proba)

    # ---- UPDATED: ROC curve line color ----
    ax1.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=2, color="C0")
    # ---------------------------------------

    ax1.plot([0, 1], [0, 1], "k--", alpha=0.5)
    ax1.set_xlabel("False Positive Rate")
    ax1.set_ylabel("True Positive Rate")
    ax1.set_title(f"ROC Curve ({prefix.upper()})")
    ax1.legend()
    ax1.grid(True, alpha=0.3)

    precision_vals, recall_vals, _ = precision_recall_curve(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    # ---- UPDATED: PR curve line color ----
    ax2.plot(recall_vals, precision_vals, label=f"AP = {pr_auc:.3f}",␣
↪linewidth=2, color="C1")
    # --------------------------------------

    ax2.set_xlabel("Recall")
    ax2.set_ylabel("Precision")
    ax2.set_title(f"Precision-Recall Curve ({prefix.upper()})")
    ax2.legend()
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_roc_pr_curves.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # Confusion matrix heatmap (threshold 0.5)
```

```
    y_pred = (y_proba >= 0.5).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
                xticklabels=["Pred 0", "Pred 1"],
                yticklabels=["True 0", "True 1"])
    plt.title(f"Confusion Matrix ({prefix.upper()})")
    plt.ylabel("True Label")
    plt.xlabel("Predicted Label")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_confusion_matrix.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()

    if cv_metrics:
        create_validation_comparison_plot(cv_metrics, output_dir, prefix)

    return imp_df


# ============================================================
# 8) SAVE RESULTS (GENERALIZED)
# ============================================================
def save_comprehensive_results(model, metrics, imp_df, df_clean, target_col,␣
 ↪feature_names, output_dir, prefix, cv_metrics=None):
    print(f"\n SAVING OUTPUTS to {output_dir}/ ...")
    os.makedirs(output_dir, exist_ok=True)

    # Metrics text file
    with open(f"{output_dir}/{prefix}_comprehensive_metrics.txt", "w",␣
 ↪encoding="utf-8") as f:
        f.write(f"COMPREHENSIVE RANDOM FOREST RESULTS ({prefix.upper()})\n")
        f.write("=" * 60 + "\n")
        f.write(f"Dataset shape: {df_clean.shape}\n")
        f.write(f"Target column: {target_col}\n")
        f.write(f"Features used: {len(feature_names)}\n\n")

        f.write("TEST SET METRICS:\n")
        for k, v in metrics.items():
            f.write(f"{k}: {v:.4f}\n")
        f.write("\n")

        if cv_metrics:
            f.write("CROSS-VALIDATION (Mean ± Std):\n")
            for metric, scores in cv_metrics.items():
```

```
                f.write(f"{metric}: {scores['test_mean']:.4f} ±␣
↪{scores['test_std']:.4f}\n")
            f.write("\n")

        f.write("TOP 10 FEATURES:\n")
        for _, row in imp_df.head(10).iterrows():
            f.write(f"{row['feature']}: {row['importance']:.6f}\n")

    # Predictions on full dataset (IMPORTANT: use the same feature columns)
    X_full = df_clean.drop(columns=[target_col], errors="ignore")
    X_full = X_full.reindex(columns=feature_names)  # align to training feature␣
↪set

    preds = model.predict_proba(X_full)[:, 1]
    predictions_df = df_clean.copy()
    predictions_df[f"{prefix}_Pred_Prob"] = preds
    predictions_df[f"{prefix}_Pred"] = (preds >= 0.5).astype(int)
    predictions_df[f"{prefix}_Correct"] = (predictions_df[f"{prefix}_Pred"] ==␣
↪predictions_df[target_col]).astype(int)

    predictions_df.to_excel(f"{output_dir}/{prefix}_predictions.xlsx",␣
↪index=False)

    # Save model
    joblib.dump(model, f"{output_dir}/{prefix}_rf_model.joblib")

    print("  Saved metrics, plots, predictions, and model.")


# ============================================================
# 9) MAIN PIPELINE (ONE FUNCTION FOR ANY BFx)
# ============================================================
def run_comprehensive_bf_analysis(
    target_bf="BF1",
    use_cross_validation=True,
    use_validation_set=True,
    df=None
):
    """
    Run the full pipeline for a given target_bf (e.g., BF1, BF3, BF4, BF5).
    If df is None, it will prompt you to upload/load.
    """
    try:
        prefix = str(target_bf).strip().lower().replace(" ", "").replace("_",␣
↪"")
        output_dir = f"{prefix}_comprehensive_outputs"
```

```
        print("  COMPREHENSIVE MICROFINANCE ANALYSIS")
        print("=" * 60)
        print(f"  Target: {target_bf}")
        print(f"  Output folder: {output_dir}/")
        print("=" * 60)

        # Load data if df not provided
        if df is None:
            try:
                import google.colab  # noqa: F401
                df = load_excel_from_colab()
            except Exception:
                df = load_excel_locally()

        print(f"  Initial dataset shape: {df.shape}")

        # Detect + preprocess target
        target_col = detect_bf_target(df, target_bf=target_bf)
        y, valid_mask = preprocess_bf_target(df[target_col],␣
↪target_label=target_col)
        df_clean = df.loc[valid_mask].copy()
        print(f"  Clean dataset: {df_clean.shape} (after removing invalid␣
↪target rows)")

        # Features
        X_final, feature_names = build_feature_matrix(df_clean, target_col)

        # CV (10-fold by default) with 6 metrics
        cv_metrics = None
        if use_cross_validation:
            cv_metrics, _ = perform_cross_validation(X_final, y, cv_folds=10)

        # Train
        model, X_train, X_test, X_val, y_train, y_test, y_val =␣
↪train_with_comprehensive_validation(
            X_final, y, use_validation_set=use_validation_set
        )

        # Evaluate test
        _, y_proba, test_metrics = evaluate_model_comprehensive(model, X_test,␣
↪y_test, "Test")

        # (Optional) Evaluate val
        if X_val is not None and y_val is not None:
            evaluate_model_comprehensive(model, X_val, y_val, "Validation")

        # Visuals
```

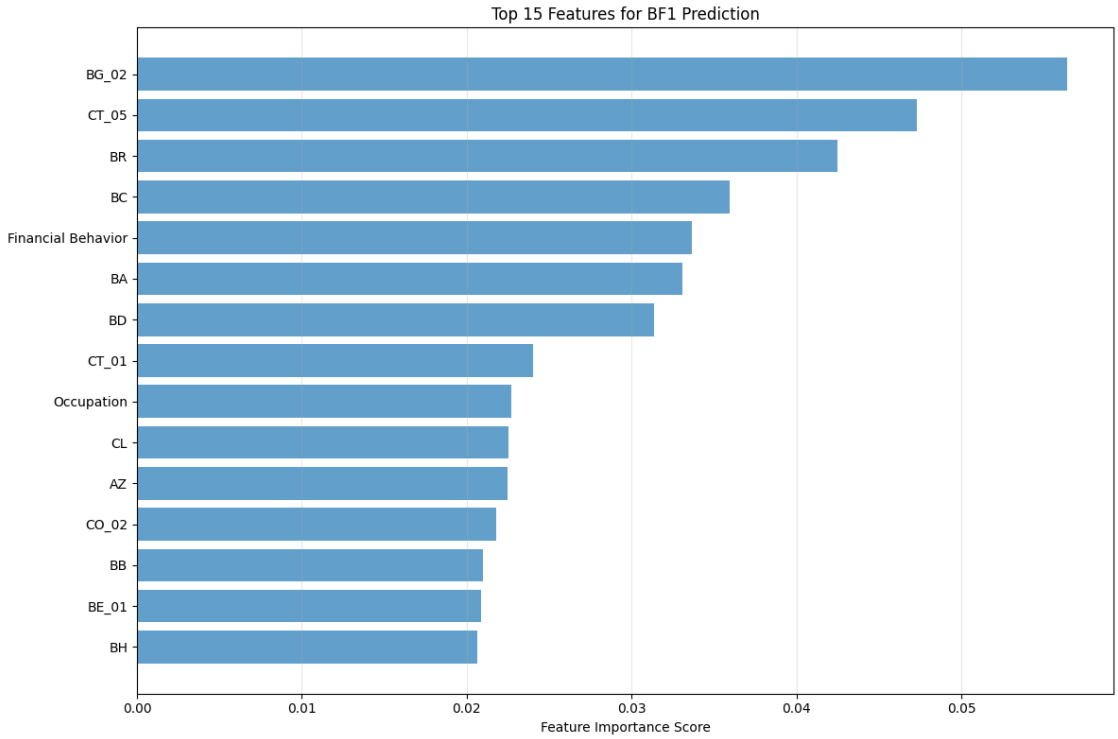
```
        imp_df = create_comprehensive_visualizations(
            y_test=y_test,
            y_proba=y_proba,
            model=model,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        # Save
        save_comprehensive_results(
            model=model,
            metrics=test_metrics,
            imp_df=imp_df,
            df_clean=df_clean,
            target_col=target_col,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        print("\n DONE.")
        return model, test_metrics, imp_df, cv_metrics

    except Exception as e:
        print(f"\n Analysis failed: {type(e).__name__}: {e}")
        import traceback
        traceback.print_exc(limit=2)
        return None, None, None, None


# ========= USAGE EXAMPLES =========
if __name__ == "__main__":
    # Single target run:
    run_comprehensive_bf_analysis(target_bf="BF1", use_cross_validation=True,␣
 ↪use_validation_set=True)

    # Multiple targets (run one by one):
    # for bf in ["BF1", "BF3", "BF4", "BF5"]:
    #     run_comprehensive_bf_analysis(target_bf=bf,␣
 ↪use_cross_validation=True, use_validation_set=True)
```

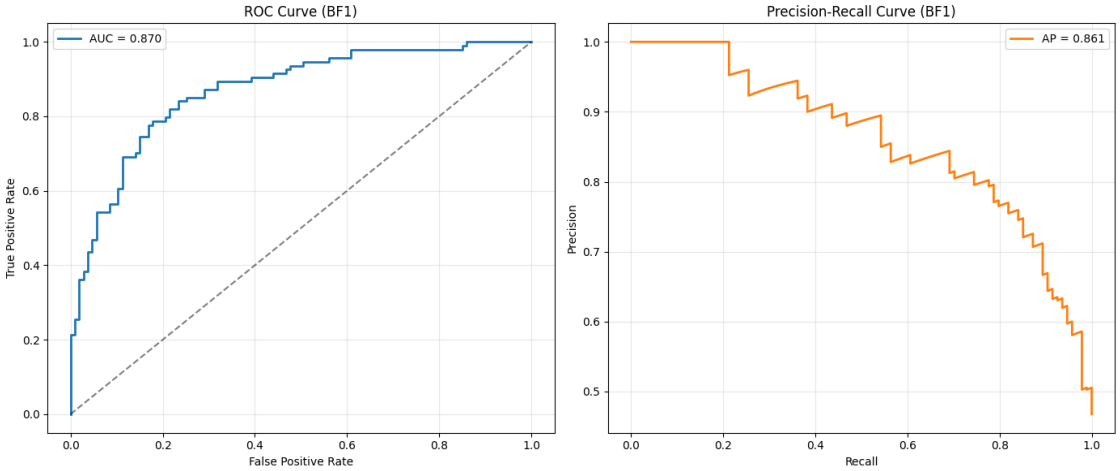
```
 COMPREHENSIVE MICROFINANCE ANALYSIS
============================================================
 Target: BF1
```

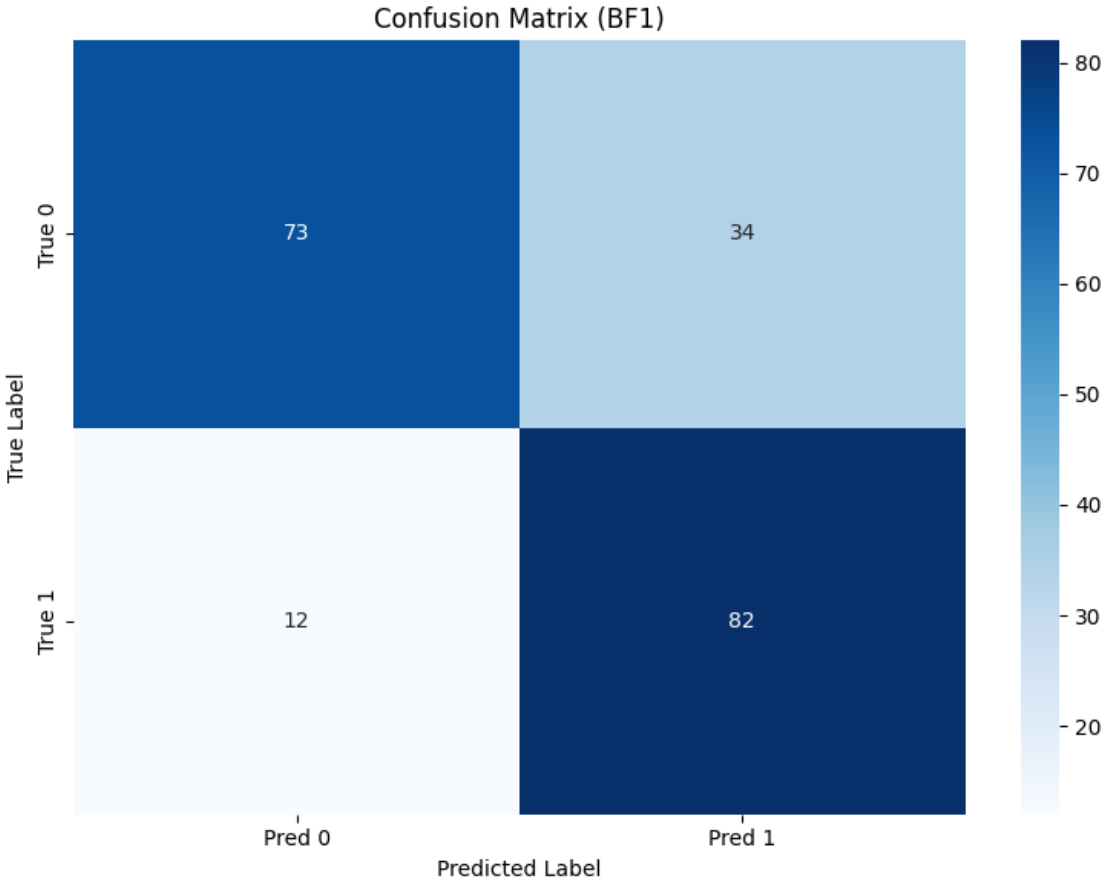
```
  Output folder: bf1_comprehensive_outputs/
============================================================
  Please select the Excel file with your encoded dataset…

<IPython.core.display.HTML object>

Saving rf_dataset_encoded_dataset.xlsx to rf_dataset_encoded_dataset.xlsx
  Loaded dataset: (1002, 85) (rows, cols)
  Initial dataset shape: (1002, 85)
  BF1 column not found automatically.
  Available columns:
['D', 'AY', 'AZ', 'BA', 'BB', 'BC', 'BD', 'BE_01', 'BE_02', 'BE_03', 'BE_04',
'BE_05', 'BE_06', 'BE_07', 'BE_08', 'BE_09', 'BE_10', 'BF', 'BG_01', 'BG_02',
'BG_03', 'BG_04', 'BG_05', 'BG_06', 'BH', 'BI', 'BN', 'BO', 'BP', 'BQ', 'BR',
'BS', 'BT', 'BU', 'BV', 'BW_01', 'BW_02', 'BW_03', 'BW_04', 'BW_05', 'BX', 'BY',
'CB', 'CD', 'CE', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM_01', 'CM_02', 'CM_03',
'CM_04', 'CM_05', 'CM_06', 'CN', 'CO_01', 'CO_02', 'CO_03', 'CO_04', 'CO_05',
'CO_06', 'CO_07', 'CO_08', 'CO_09', 'CP', 'CR_01', 'CR_02', 'CR_03', 'CR_04',
'CS', 'CT_01', 'CT_02', 'CT_03', 'CT_04', 'CT_05', 'CT_06', 'CU', 'Occupation',
'Financial Behavior', 'Family', 'Lifestyle', 'Digital Literacy']
  Enter exact column name for BF1: BG_01
  Using specified target column: BG_01

  TARGET ANALYSIS: BG_01
----------------------------------------
Value distribution:
BG_01
0    533
1    469
Name: count, dtype: int64
Unique values: [np.int64(0), np.int64(1)]
  Final class balance: {0: 0.532, 1: 0.468}
  Clean dataset: (1002, 85) (after removing invalid target rows)

  FEATURE ENGINEERING
----------------------------------------
  Found 84 numeric features
  Handling missing values with median imputation…
  Final feature matrix: (1002, 84) (rows, cols)

  PERFORMING 10-FOLD CROSS-VALIDATION
==================================================
  CROSS-VALIDATION RESULTS (Mean ± Std):
----------------------------------------
ACCURACY       : 0.7943 ± 0.0413 (train: 0.9055)
PRECISION      : 0.7376 ± 0.0379 (train: 0.8544)
RECALL         : 0.8722 ± 0.0484 (train: 0.9623)
F1             : 0.7989 ± 0.0393 (train: 0.9051)
ROC_AUC        : 0.8911 ± 0.0344 (train: 0.9762)
```

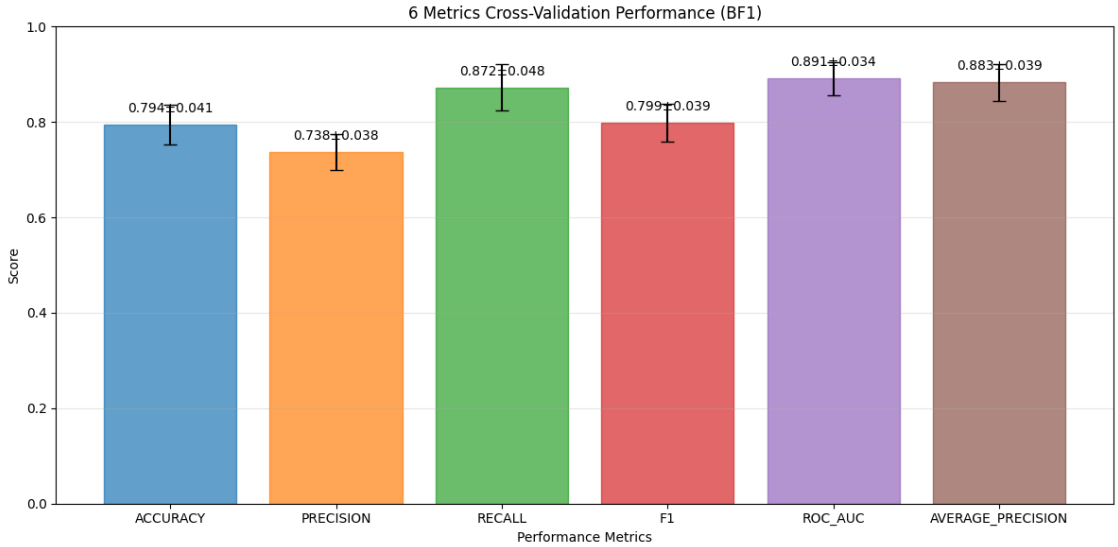
```
AVERAGE_PRECISION: 0.8832 ± 0.0390 (train: 0.9757)

  Overfitting Analysis:
    Train-Test Accuracy Gap: 0.1112
       Potential overfitting detected

  MODEL TRAINING: Random Forest
----------------------------------------

  CREATING VALIDATION SET (20% of data)
==================================================
  Training set: 600 samples
  Test set: 201 samples
  Validation set: 201 samples
  Training model with regularization…

============================================================
  TEST SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.7711
  Precision: 0.7069
  Recall:    0.8723
  F1-Score:  0.7810
  ROC-AUC:   0.8699
  PR-AUC:    0.8609

  Test Classification Report:
              precision    recall  f1-score   support

           0      0.859     0.682     0.760       107
           1      0.707     0.872     0.781        94

    accuracy                          0.771       201
   macro avg      0.783     0.777     0.771       201
weighted avg      0.788     0.771     0.770       201

  Test Confusion Matrix:
[[TN FP]
 [FN TP]]
[[73 34]
 [12 82]]

============================================================
  VALIDATION SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.7960
  Precision: 0.7387
  Recall:    0.8723
```

```
 F1-Score:  0.8000
 ROC-AUC:   0.8913
 PR-AUC:    0.8924

 Validation Classification Report:
              precision    recall  f1-score   support

           0      0.867     0.729     0.792       107
           1      0.739     0.872     0.800        94

    accuracy                         0.796       201
   macro avg      0.803     0.801     0.796       201
weighted avg      0.807     0.796     0.796       201

 Validation Confusion Matrix:
[[TN FP]
 [FN TP]]
[[78 29]
 [12 82]]
```




ROC Curve (BF1)
AUC = 0.870
True Positive Rate
False Positive Rate
Precision-Recall Curve (BF1)
AP = 0.861
Precision
Recall


Confusion Matrix (BF1)
True 0
True 1
True Label
73
34
12
82
Pred 0
Pred 1
Predicted Label




```
SAVING OUTPUTS to bf1_comprehensive_outputs/ …
Saved metrics, plots, predictions, and model.

DONE.
```

[2]:
```
"""
Reusable Random Forest Pipeline for Microfinance Impact Analysis
----------------------------------------------------------------
Use this to predict BF1 / BF3 / BF4 / BF5 ... by changing ONE variable.

Example:
    run_comprehensive_bf_analysis(target_bf="BF1")
    run_comprehensive_bf_analysis(target_bf="BF3")
"""

# ===========
# 0) IMPORTS
# ===========
import os, io, re, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪cross_validate
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    accuracy_score, precision_score, recall_score, f1_score
)
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
import joblib

RANDOM_STATE = 42


# ============================================================
# 1) DATA LOADING
# ============================================================
def load_excel_from_colab():
    """Upload Excel in Google Colab."""
    try:
        from google.colab import files
        print(" Please select the Excel file with your encoded dataset...")
        uploaded = files.upload()
        path = list(uploaded.keys())[0]
        df = pd.read_excel(path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        raise RuntimeError("Colab upload failed.") from e


def load_excel_locally():
    """Upload Excel when running locally."""
    try:
        import tkinter as tk
        from tkinter import filedialog
        print(" Please select the Excel file...")
        root = tk.Tk()
        root.withdraw()
        file_path = filedialog.askopenfilename(
            title="Select Excel file",
            filetypes=[("Excel files", "*.xlsx *.xls")]
        )
        if not file_path:
            raise ValueError("No file selected")
        df = pd.read_excel(file_path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
```

```
        return df
    except Exception as e:
        print(f" File dialog failed: {e}")
        file_path = input(" Enter full path to your Excel file: ")
        df = pd.read_excel(file_path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
        return df


# ============================================================
# 2) TARGET SELECTION (GENERALIZED BFx)
# ============================================================
def detect_bf_target(df: pd.DataFrame, target_bf: str = "BF1") -> str:
    """
    Detect BF target column robustly.
    target_bf examples: "BF1", "BF3", "BF5"
    """
    m = re.match(r"^\s*BF\s*(\d+)\s*$", str(target_bf), flags=re.I)
    if not m:
        raise ValueError(f"target_bf must look like 'BF1', 'BF3', etc. Got:␣
↪{target_bf}")

    k = m.group(1)  # number as string
    aliases = [
        f"BF{k}", f"bf{k}",
        f"BF {k}", f"bf {k}",
        f"BF_{k}", f"bf_{k}",
        f"BF-{k}", f"bf-{k}",
    ]

    for a in aliases:
        if a in df.columns:
            print(f" Found target column: {a}")
            return a

    # case-insensitive regex search in column names
    pat = re.compile(rf"\bbf\s*[_-]?\s*{k}\b", flags=re.I)
    for col in df.columns:
        if pat.search(str(col)):
            print(f" Found target column (regex match): {col}")
            return col

    print(f" {target_bf} column not found automatically.")
    print(" Available columns:")
    print(df.columns.tolist())
    user_col = input(f" Enter exact column name for {target_bf}: ").strip()
    if user_col in df.columns:
```

```
        print(f"  Using specified target column: {user_col}")
        return user_col
    raise ValueError(f"Column '{user_col}' not found in dataset")


def preprocess_bf_target(y_series: pd.Series, target_label: str):
    """
    Convert BF target responses into binary classification (expects 0/1 after␣
 ↪cleaning).
    """
    y_numeric = pd.to_numeric(y_series, errors="coerce")
    valid_mask = y_numeric.notna() & np.isfinite(y_numeric)
    y_clean = y_numeric[valid_mask]

    if len(y_clean) == 0:
        raise ValueError(f"  No valid values for {target_label} after cleaning")

    print(f"\n  TARGET ANALYSIS: {target_label}")
    print("-" * 40)
    print("Value distribution:")
    print(y_clean.value_counts().sort_index())

    unique_vals = sorted(y_clean.unique())
    print(f"Unique values: {unique_vals}")

    # If your encoded dataset is already 0/1, this is perfect.
    # If it includes 1/2 or 1/3 scales, you should define a threshold rule here.
    y_binary = y_clean.astype(int)

    class_balance = y_binary.value_counts(normalize=True).round(3)
    print(f"  Final class balance: {class_balance.to_dict()}")

    return y_binary, valid_mask


# ============================================================
# 3) FEATURE ENGINEERING
# ============================================================
def build_feature_matrix(df_clean: pd.DataFrame, target_col: str):
    """Prepare features for ML (numeric-only)."""
    print("\n  FEATURE ENGINEERING")
    print("-" * 40)

    X = df_clean.drop(columns=[target_col], errors="ignore")
    numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
    print(f"  Found {len(numeric_cols)} numeric features")
```

```
    if len(numeric_cols) == 0:
        raise ValueError("  No numeric features found for modeling")

    print("  Handling missing values with median imputation...")
    imputer = SimpleImputer(strategy="median")
    X_imputed = imputer.fit_transform(X[numeric_cols])
    X_imputed = pd.DataFrame(X_imputed, columns=numeric_cols, index=X.index)

    non_constant_cols = [c for c in X_imputed.columns if X_imputed[c].nunique()␣
 ↪> 1]
    dropped = len(numeric_cols) - len(non_constant_cols)
    if dropped > 0:
        print(f"   Removed {dropped} constant columns")

    X_final = X_imputed[non_constant_cols]
    print(f"  Final feature matrix: {X_final.shape} (rows, cols)")
    return X_final, non_constant_cols


# ============================================================
# 4) CROSS-VALIDATION
# ============================================================
def perform_cross_validation(X, y, cv_folds=10):
    print(f"\n  PERFORMING {cv_folds}-FOLD CROSS-VALIDATION")
    print("=" * 50)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    scoring = {
        "accuracy": "accuracy",
        "precision": "precision",
```

```
        "recall": "recall",
        "f1": "f1",
        "roc_auc": "roc_auc",
        "average_precision": "average_precision"
    }

    cv_results = cross_validate(
        pipe, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,␣
↪random_state=RANDOM_STATE),
        scoring=scoring,
        return_train_score=True,
        n_jobs=-1
    )

    print(" CROSS-VALIDATION RESULTS (Mean ± Std):")
    print("-" * 40)

    metrics_summary = {}
    for metric in scoring.keys():
        test_scores = cv_results[f"test_{metric}"]
        train_scores = cv_results[f"train_{metric}"]

        test_mean = float(np.mean(test_scores))
        test_std = float(np.std(test_scores))
        train_mean = float(np.mean(train_scores))

        metrics_summary[metric] = {
            "test_mean": test_mean,
            "test_std": test_std,
            "train_mean": train_mean
        }

        print(f"{metric.upper():<15}: {test_mean:.4f} ± {test_std:.4f} (train:␣
↪{train_mean:.4f})")

    overfit_gap = metrics_summary["accuracy"]["train_mean"] -␣
↪metrics_summary["accuracy"]["test_mean"]
    print(f"\n Overfitting Analysis:")
    print(f"   Train-Test Accuracy Gap: {overfit_gap:.4f}")
    print("     Potential overfitting detected" if overfit_gap > 0.05 else "  ␣
↪ Model generalizes well")

    return metrics_summary, cv_results


# ============================================================
```

```
# 5) TRAIN/TEST/VALIDATION SPLITS + TRAINING
# ============================================================
def create_validation_set(X, y, validation_size=0.2):
    print(f"\n CREATING VALIDATION SET ({validation_size*100:.0f}% of data)")
    print("=" * 50)

    X_train_test, X_val, y_train_test, y_val = train_test_split(
        X, y, test_size=validation_size, stratify=y, random_state=RANDOM_STATE
    )

    X_train, X_test, y_train, y_test = train_test_split(
        X_train_test, y_train_test, test_size=0.25, stratify=y_train_test,␣
 ↪random_state=RANDOM_STATE
    )

    print(f" Training set: {X_train.shape[0]} samples")
    print(f" Test set: {X_test.shape[0]} samples")
    print(f" Validation set: {X_val.shape[0]} samples")

    return X_train, X_test, X_val, y_train, y_test, y_val


def train_with_comprehensive_validation(X, y, use_validation_set=True):
    print("\n MODEL TRAINING: Random Forest")
    print("-" * 40)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    if use_validation_set:
        X_train, X_test, X_val, y_train, y_test, y_val =␣
 ↪create_validation_set(X, y)
```

```
        print("  Training model with regularization...")
        pipe.fit(X_train, y_train)
        return pipe, X_train, X_test, X_val, y_train, y_test, y_val

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, stratify=y, random_state=RANDOM_STATE
    )
    print(f"  Training set: {X_train.shape[0]} samples")
    print(f"  Test set: {X_test.shape[0]} samples")
    print("  Training model with regularization...")
    pipe.fit(X_train, y_train)
    return pipe, X_train, X_test, None, y_train, y_test, None


# ============================================================
# 6) EVALUATION
# ============================================================
def evaluate_model_comprehensive(model, X_test, y_test, dataset_name="Test"):
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    print("\n" + "=" * 60)
    print(f"  {dataset_name.upper()} SET COMPREHENSIVE EVALUATION")
    print("=" * 60)
    print(f"  Accuracy:  {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall:    {rec:.4f}")
    print(f"  F1-Score:  {f1:.4f}")
    print(f"  ROC-AUC:   {roc_auc:.4f}")
    print(f"  PR-AUC:    {pr_auc:.4f}")

    print(f"\n {dataset_name} Classification Report:")
    print(classification_report(y_test, y_pred, digits=3))

    cm = confusion_matrix(y_test, y_pred)
    print(f"  {dataset_name} Confusion Matrix:")
    print("[[TN FP]\n [FN TP]]")
    print(cm)

    return y_pred, y_proba, {
```

```
        "accuracy": acc, "precision": prec, "recall": rec, "f1": f1,
        "roc_auc": roc_auc, "pr_auc": pr_auc
    }


# ============================================================
# 7) VISUALIZATIONS (UPDATED COLORS ADDED)
# ============================================================
def create_validation_comparison_plot(cv_metrics, output_dir, prefix):
    metrics_names = list(cv_metrics.keys())
    test_means = [cv_metrics[m]["test_mean"] for m in metrics_names]
    test_stds = [cv_metrics[m]["test_std"] for m in metrics_names]

    plt.figure(figsize=(12, 6))
    x_pos = np.arange(len(metrics_names))

    # ---- UPDATED: bars have different colors automatically ----
    bars = plt.bar(
        x_pos,
        test_means,
        yerr=test_stds,
        capsize=5,
        alpha=0.7
    )
    for i, b in enumerate(bars):
        b.set_color(f"C{i}")
    # ------------------------------------------------------------

    plt.xlabel("Performance Metrics")
    plt.ylabel("Score")
    plt.title(f"{len(metrics_names)} Metrics Cross-Validation Performance␣
↪({prefix.upper()})")
    plt.xticks(x_pos, [m.upper() for m in metrics_names], rotation=0)
    plt.ylim(0, 1)
    plt.grid(True, alpha=0.3, axis="y")

    for i, (mean, std) in enumerate(zip(test_means, test_stds)):
        plt.text(i, mean + 0.02, f"{mean:.3f}±{std:.3f}", ha="center",␣
↪va="bottom", fontsize=10)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_cross_validation_results.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()
```

```
def create_comprehensive_visualizations(y_test, y_proba, model, feature_names,␣
 ↪output_dir, prefix, cv_metrics=None):
    os.makedirs(output_dir, exist_ok=True)

    rf_model = model.named_steps["rf"]
    imp_df = pd.DataFrame({
        "feature": feature_names,
        "importance": rf_model.feature_importances_
    }).sort_values("importance", ascending=False)

    imp_df.to_csv(f"{output_dir}/{prefix}_feature_importances.csv", index=False)

    # Feature importance plot
    plt.figure(figsize=(12, 8))
    top_features = imp_df.head(15).iloc[::-1]
    plt.barh(range(len(top_features)), top_features["importance"], alpha=0.7)
    plt.yticks(range(len(top_features)), top_features["feature"])
    plt.xlabel("Feature Importance Score")
    plt.title(f"Top 15 Features for {prefix.upper()} Prediction")
    plt.grid(True, alpha=0.3, axis="x")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_feature_importances.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()

    # ROC + PR curves
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    roc_auc = roc_auc_score(y_test, y_proba)

    # ---- UPDATED: ROC curve line color ----
    ax1.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=2, color="C0")
    # ---------------------------------------

    ax1.plot([0, 1], [0, 1], "k--", alpha=0.5)
    ax1.set_xlabel("False Positive Rate")
    ax1.set_ylabel("True Positive Rate")
    ax1.set_title(f"ROC Curve ({prefix.upper()})")
    ax1.legend()
    ax1.grid(True, alpha=0.3)

    precision_vals, recall_vals, _ = precision_recall_curve(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    # ---- UPDATED: PR curve line color ----
```

```
    ax2.plot(recall_vals, precision_vals, label=f"AP = {pr_auc:.3f}",␣
↪linewidth=2, color="C1")
    # ------------------------------------

    ax2.set_xlabel("Recall")
    ax2.set_ylabel("Precision")
    ax2.set_title(f"Precision-Recall Curve ({prefix.upper()})")
    ax2.legend()
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_roc_pr_curves.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # Confusion matrix heatmap (threshold 0.5)
    y_pred = (y_proba >= 0.5).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
                xticklabels=["Pred 0", "Pred 1"],
                yticklabels=["True 0", "True 1"])
    plt.title(f"Confusion Matrix ({prefix.upper()})")
    plt.ylabel("True Label")
    plt.xlabel("Predicted Label")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_confusion_matrix.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    if cv_metrics:
        create_validation_comparison_plot(cv_metrics, output_dir, prefix)

    return imp_df


# ============================================================
# 8) SAVE RESULTS (GENERALIZED)
# ============================================================
def save_comprehensive_results(model, metrics, imp_df, df_clean, target_col,␣
↪feature_names, output_dir, prefix, cv_metrics=None):
    print(f"\n SAVING OUTPUTS to {output_dir}/ ...")
    os.makedirs(output_dir, exist_ok=True)

    # Metrics text file
```

```
    with open(f"{output_dir}/{prefix}_comprehensive_metrics.txt", "w",␣
↪encoding="utf-8") as f:
        f.write(f"COMPREHENSIVE RANDOM FOREST RESULTS ({prefix.upper()})\n")
        f.write("=" * 60 + "\n")
        f.write(f"Dataset shape: {df_clean.shape}\n")
        f.write(f"Target column: {target_col}\n")
        f.write(f"Features used: {len(feature_names)}\n\n")

        f.write("TEST SET METRICS:\n")
        for k, v in metrics.items():
            f.write(f"{k}: {v:.4f}\n")
        f.write("\n")

        if cv_metrics:
            f.write("CROSS-VALIDATION (Mean ± Std):\n")
            for metric, scores in cv_metrics.items():
                f.write(f"{metric}: {scores['test_mean']:.4f} ±␣
↪{scores['test_std']:.4f}\n")
            f.write("\n")

        f.write("TOP 10 FEATURES:\n")
        for _, row in imp_df.head(10).iterrows():
            f.write(f"{row['feature']}: {row['importance']:.6f}\n")

    # Predictions on full dataset (IMPORTANT: use the same feature columns)
    X_full = df_clean.drop(columns=[target_col], errors="ignore")
    X_full = X_full.reindex(columns=feature_names)  # align to training feature␣
↪set

    preds = model.predict_proba(X_full)[:, 1]
    predictions_df = df_clean.copy()
    predictions_df[f"{prefix}_Pred_Prob"] = preds
    predictions_df[f"{prefix}_Pred"] = (preds >= 0.5).astype(int)
    predictions_df[f"{prefix}_Correct"] = (predictions_df[f"{prefix}_Pred"] ==␣
↪predictions_df[target_col]).astype(int)

    predictions_df.to_excel(f"{output_dir}/{prefix}_predictions.xlsx",␣
↪index=False)

    # Save model
    joblib.dump(model, f"{output_dir}/{prefix}_rf_model.joblib")

    print("  Saved metrics, plots, predictions, and model.")


# ============================================================
# 9) MAIN PIPELINE (ONE FUNCTION FOR ANY BFx)
```

```python
# ============================================================
def run_comprehensive_bf_analysis(
    target_bf="BF1",
    use_cross_validation=True,
    use_validation_set=True,
    df=None
):
    """
    Run the full pipeline for a given target_bf (e.g., BF1, BF3, BF4, BF5).
    If df is None, it will prompt you to upload/load.
    """
    try:
        prefix = str(target_bf).strip().lower().replace(" ", "").replace("_",␣
 ↪"")
        output_dir = f"{prefix}_comprehensive_outputs"

        print(" COMPREHENSIVE MICROFINANCE ANALYSIS")
        print("=" * 60)
        print(f" Target: {target_bf}")
        print(f" Output folder: {output_dir}/")
        print("=" * 60)

        # Load data if df not provided
        if df is None:
            try:
                import google.colab  # noqa: F401
                df = load_excel_from_colab()
            except Exception:
                df = load_excel_locally()

        print(f" Initial dataset shape: {df.shape}")

        # Detect + preprocess target
        target_col = detect_bf_target(df, target_bf=target_bf)
        y, valid_mask = preprocess_bf_target(df[target_col],␣
 ↪target_label=target_col)
        df_clean = df.loc[valid_mask].copy()
        print(f" Clean dataset: {df_clean.shape} (after removing invalid␣
 ↪target rows)")

        # Features
        X_final, feature_names = build_feature_matrix(df_clean, target_col)

        # CV (10-fold by default) with 6 metrics
        cv_metrics = None
        if use_cross_validation:
            cv_metrics, _ = perform_cross_validation(X_final, y, cv_folds=10)
```

```
        # Train
        model, X_train, X_test, X_val, y_train, y_test, y_val =␣
↪train_with_comprehensive_validation(
            X_final, y, use_validation_set=use_validation_set
        )

        # Evaluate test
        _, y_proba, test_metrics = evaluate_model_comprehensive(model, X_test,␣
↪y_test, "Test")

        # (Optional) Evaluate val
        if X_val is not None and y_val is not None:
            evaluate_model_comprehensive(model, X_val, y_val, "Validation")

        # Visuals
        imp_df = create_comprehensive_visualizations(
            y_test=y_test,
            y_proba=y_proba,
            model=model,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        # Save
        save_comprehensive_results(
            model=model,
            metrics=test_metrics,
            imp_df=imp_df,
            df_clean=df_clean,
            target_col=target_col,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        print("\n DONE.")
        return model, test_metrics, imp_df, cv_metrics

    except Exception as e:
        print(f"\n Analysis failed: {type(e).__name__}: {e}")
        import traceback
        traceback.print_exc(limit=2)
        return None, None, None, None
```

```python
# ========= USAGE EXAMPLES =========
if __name__ == "__main__":
    # Single target run:
    run_comprehensive_bf_analysis(target_bf="BF3", use_cross_validation=True,␣
 ↪use_validation_set=True)

    # Multiple targets (run one by one):
    # for bf in ["BF1", "BF3", "BF4", "BF5"]:
    #     run_comprehensive_bf_analysis(target_bf=bf,␣
 ↪use_cross_validation=True, use_validation_set=True)
```

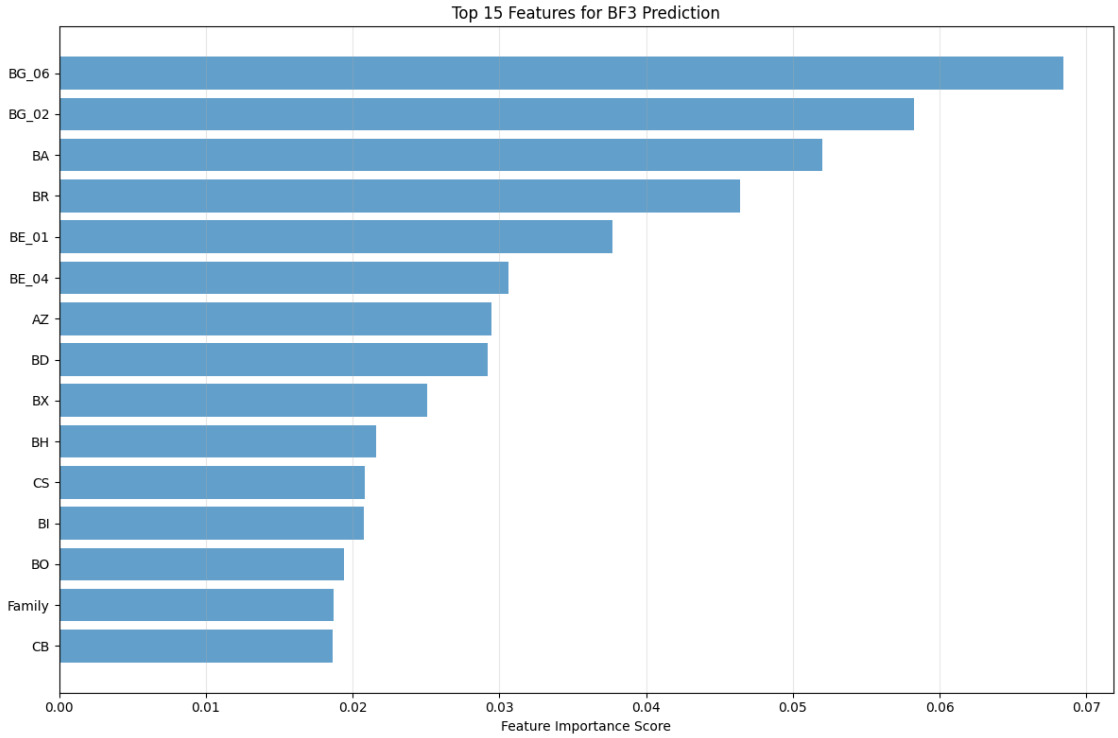
```
 COMPREHENSIVE MICROFINANCE ANALYSIS
============================================================
 Target: BF3
 Output folder: bf3_comprehensive_outputs/
============================================================
 Please select the Excel file with your encoded dataset…
```

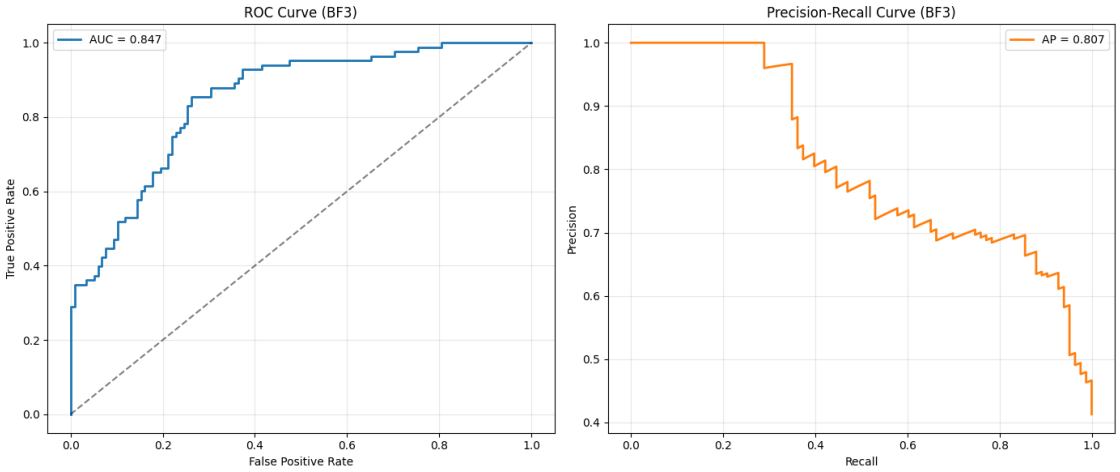
```
<IPython.core.display.HTML object>
```

```
Saving rf_dataset_encoded_dataset.xlsx to rf_dataset_encoded_dataset (1).xlsx
 Loaded dataset: (1002, 85) (rows, cols)
 Initial dataset shape: (1002, 85)
 BF3 column not found automatically.
 Available columns:
['D', 'AY', 'AZ', 'BA', 'BB', 'BC', 'BD', 'BE_01', 'BE_02', 'BE_03', 'BE_04',
'BE_05', 'BE_06', 'BE_07', 'BE_08', 'BE_09', 'BE_10', 'BF', 'BG_01', 'BG_02',
'BG_03', 'BG_04', 'BG_05', 'BG_06', 'BH', 'BI', 'BN', 'BO', 'BP', 'BQ', 'BR',
'BS', 'BT', 'BU', 'BV', 'BW_01', 'BW_02', 'BW_03', 'BW_04', 'BW_05', 'BX', 'BY',
'CB', 'CD', 'CE', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM_01', 'CM_02', 'CM_03',
'CM_04', 'CM_05', 'CM_06', 'CN', 'CO_01', 'CO_02', 'CO_03', 'CO_04', 'CO_05',
'CO_06', 'CO_07', 'CO_08', 'CO_09', 'CP', 'CR_01', 'CR_02', 'CR_03', 'CR_04',
'CS', 'CT_01', 'CT_02', 'CT_03', 'CT_04', 'CT_05', 'CT_06', 'CU', 'Occupation',
'Financial Behavior', 'Family', 'Lifestyle', 'Digital Literacy']
 Enter exact column name for BF3: BG_03
 Using specified target column: BG_03

 TARGET ANALYSIS: BG_03
----------------------------------------
Value distribution:
BG_03
0    590
1    412
Name: count, dtype: int64
Unique values: [np.int64(0), np.int64(1)]
 Final class balance: {0: 0.589, 1: 0.411}
```

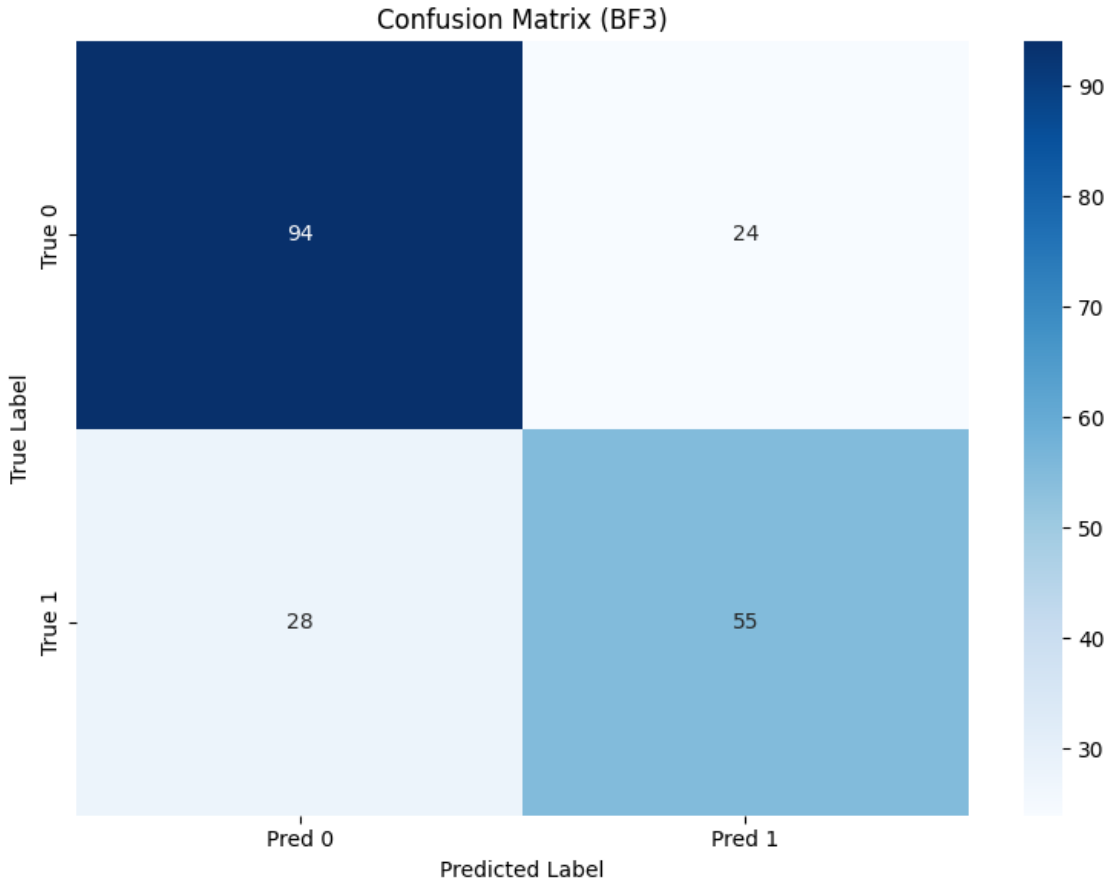
```
  Clean dataset: (1002, 85) (after removing invalid target rows)

  FEATURE ENGINEERING
----------------------------------------
  Found 84 numeric features
  Handling missing values with median imputation…
  Final feature matrix: (1002, 84) (rows, cols)

  PERFORMING 10-FOLD CROSS-VALIDATION
==================================================
  CROSS-VALIDATION RESULTS (Mean ± Std):
----------------------------------------
ACCURACY       : 0.8244 ± 0.0360 (train: 0.9189)
PRECISION      : 0.7840 ± 0.0524 (train: 0.8919)
RECALL         : 0.7961 ± 0.0525 (train: 0.9137)
F1             : 0.7886 ± 0.0413 (train: 0.9026)
ROC_AUC        : 0.9034 ± 0.0236 (train: 0.9801)
AVERAGE_PRECISION: 0.8776 ± 0.0292 (train: 0.9723)

  Overfitting Analysis:
    Train-Test Accuracy Gap: 0.0945
       Potential overfitting detected

  MODEL TRAINING: Random Forest
----------------------------------------

  CREATING VALIDATION SET (20% of data)
==================================================
  Training set: 600 samples
  Test set: 201 samples
  Validation set: 201 samples
  Training model with regularization…

============================================================
  TEST SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.7413
  Precision: 0.6962
  Recall:    0.6627
  F1-Score:  0.6790
  ROC-AUC:   0.8474
  PR-AUC:    0.8069

  Test Classification Report:
              precision    recall  f1-score   support

           0      0.770     0.797     0.783       118
           1      0.696     0.663     0.679        83
```

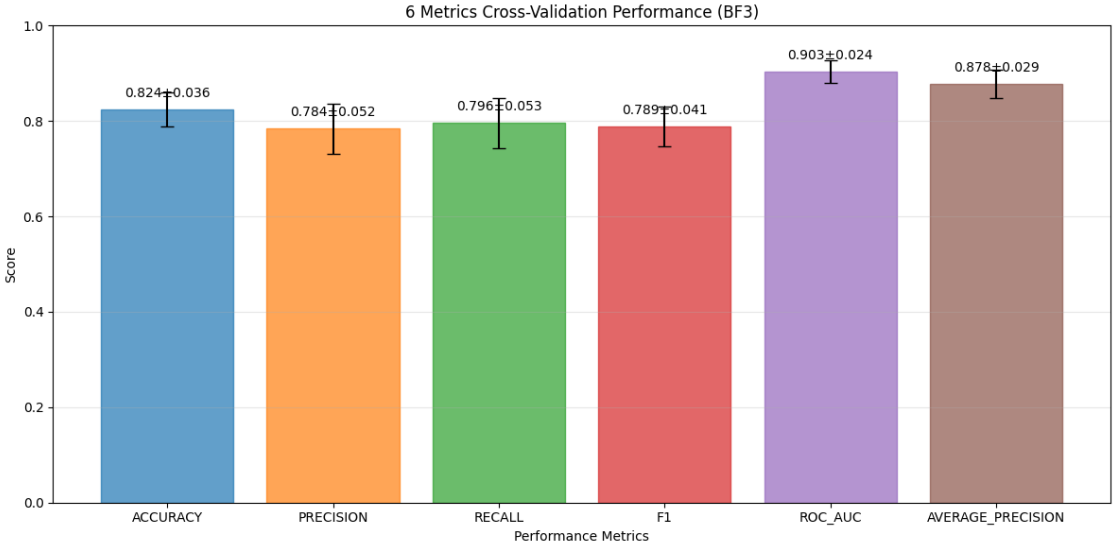
```
    accuracy                           0.741       201
   macro avg       0.733     0.730     0.731       201
weighted avg       0.740     0.741     0.740       201

  Test Confusion Matrix:
[[TN FP]
 [FN TP]]
[[94 24]
 [28 55]]

============================================================
  VALIDATION SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.7910
  Precision: 0.7595
  Recall:    0.7229
  F1-Score:  0.7407
  ROC-AUC:   0.8703
  PR-AUC:    0.8409

  Validation Classification Report:
              precision    recall  f1-score   support

           0       0.811     0.839     0.825       118
           1       0.759     0.723     0.741        83

    accuracy                           0.791       201
   macro avg       0.785     0.781     0.783       201
weighted avg       0.790     0.791     0.790       201

  Validation Confusion Matrix:
[[TN FP]
 [FN TP]]
[[99 19]
 [23 60]]
```


Top 15 Features for BF3 Prediction
BG_06
BG_02
BA
BR
BE_01
BE_04
AZ
BD
BX
BH
CS
BI
BO
Family
CB
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Feature Importance Score


ROC Curve (BF3)
AUC = 0.847
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0

Precision-Recall Curve (BF3)
AP = 0.807
Precision
Recall
0.4
0.5
0.6
0.7
0.8
0.9
1.0


Confusion Matrix (BF3)
True 0
True 1
True Label
94
24
28
55
Pred 0
Pred 1
Predicted Label
90
80
70
60
50
40
30


6 Metrics Cross-Validation Performance (BF3)
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.824±0.036
0.784±0.052
0.796±0.053
0.789±0.041
0.903±0.024
0.878±0.029
ACCURACY
PRECISION
RECALL
F1
ROC_AUC
AVERAGE_PRECISION
Performance Metrics

```
SAVING OUTPUTS to bf3_comprehensive_outputs/ …
Saved metrics, plots, predictions, and model.

DONE.
```

[3]:
```
"""
Reusable Random Forest Pipeline for Microfinance Impact Analysis
----------------------------------------------------------------
Use this to predict BF1 / BF3 / BF4 / BF5 ... by changing ONE variable.

Example:
    run_comprehensive_bf_analysis(target_bf="BF1")
    run_comprehensive_bf_analysis(target_bf="BF3")
"""

# ===========
# 0) IMPORTS
# ===========
import os, io, re, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪cross_validate
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    accuracy_score, precision_score, recall_score, f1_score
)
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
import joblib

RANDOM_STATE = 42


# ============================================================
# 1) DATA LOADING
# ============================================================
def load_excel_from_colab():
    """Upload Excel in Google Colab."""
```

```
    try:
        from google.colab import files
        print("  Please select the Excel file with your encoded dataset...")
        uploaded = files.upload()
        path = list(uploaded.keys())[0]
        df = pd.read_excel(path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        raise RuntimeError("Colab upload failed.") from e


def load_excel_locally():
    """Upload Excel when running locally."""
    try:
        import tkinter as tk
        from tkinter import filedialog
        print("  Please select the Excel file...")
        root = tk.Tk()
        root.withdraw()
        file_path = filedialog.askopenfilename(
            title="Select Excel file",
            filetypes=[("Excel files", "*.xlsx *.xls")]
        )
        if not file_path:
            raise ValueError("No file selected")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        print(f"  File dialog failed: {e}")
        file_path = input("  Enter full path to your Excel file: ")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df


# ============================================================
# 2) TARGET SELECTION (GENERALIZED BFx)
# ============================================================
def detect_bf_target(df: pd.DataFrame, target_bf: str = "BF1") -> str:
    """
    Detect BF target column robustly.
    target_bf examples: "BF1", "BF3", "BF5"
    """
    m = re.match(r"^\s*BF\s*(\d+)\s*$", str(target_bf), flags=re.I)
    if not m:
```

```
        raise ValueError(f"target_bf must look like 'BF1', 'BF3', etc. Got:␣
↪{target_bf}")

    k = m.group(1)  # number as string
    aliases = [
        f"BF{k}", f"bf{k}",
        f"BF {k}", f"bf {k}",
        f"BF_{k}", f"bf_{k}",
        f"BF-{k}", f"bf-{k}",
    ]

    for a in aliases:
        if a in df.columns:
            print(f"  Found target column: {a}")
            return a

    # case-insensitive regex search in column names
    pat = re.compile(rf"\bbf\s*[_-]?\s*{k}\b", flags=re.I)
    for col in df.columns:
        if pat.search(str(col)):
            print(f"  Found target column (regex match): {col}")
            return col

    print(f"  {target_bf} column not found automatically.")
    print("  Available columns:")
    print(df.columns.tolist())
    user_col = input(f"  Enter exact column name for {target_bf}: ").strip()
    if user_col in df.columns:
        print(f"  Using specified target column: {user_col}")
        return user_col
    raise ValueError(f"Column '{user_col}' not found in dataset")


def preprocess_bf_target(y_series: pd.Series, target_label: str):
    """
    Convert BF target responses into binary classification (expects 0/1 after␣
↪cleaning).
    """
    y_numeric = pd.to_numeric(y_series, errors="coerce")
    valid_mask = y_numeric.notna() & np.isfinite(y_numeric)
    y_clean = y_numeric[valid_mask]

    if len(y_clean) == 0:
        raise ValueError(f"  No valid values for {target_label} after cleaning")

    print(f"\n  TARGET ANALYSIS: {target_label}")
    print("-" * 40)
```

```
    print("Value distribution:")
    print(y_clean.value_counts().sort_index())

    unique_vals = sorted(y_clean.unique())
    print(f"Unique values: {unique_vals}")

    # If your encoded dataset is already 0/1, this is perfect.
    # If it includes 1/2 or 1/3 scales, you should define a threshold rule here.
    y_binary = y_clean.astype(int)

    class_balance = y_binary.value_counts(normalize=True).round(3)
    print(f"  Final class balance: {class_balance.to_dict()}")

    return y_binary, valid_mask


# ============================================================
# 3) FEATURE ENGINEERING
# ============================================================
def build_feature_matrix(df_clean: pd.DataFrame, target_col: str):
    """Prepare features for ML (numeric-only)."""
    print("\n FEATURE ENGINEERING")
    print("-" * 40)

    X = df_clean.drop(columns=[target_col], errors="ignore")
    numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
    print(f"  Found {len(numeric_cols)} numeric features")

    if len(numeric_cols) == 0:
        raise ValueError("  No numeric features found for modeling")

    print("  Handling missing values with median imputation...")
    imputer = SimpleImputer(strategy="median")
    X_imputed = imputer.fit_transform(X[numeric_cols])
    X_imputed = pd.DataFrame(X_imputed, columns=numeric_cols, index=X.index)

    non_constant_cols = [c for c in X_imputed.columns if X_imputed[c].nunique()␣
 ↪> 1]
    dropped = len(numeric_cols) - len(non_constant_cols)
    if dropped > 0:
        print(f"   Removed {dropped} constant columns")

    X_final = X_imputed[non_constant_cols]
    print(f"  Final feature matrix: {X_final.shape} (rows, cols)")
    return X_final, non_constant_cols
```

```python
# ============================================================
# 4) CROSS-VALIDATION
# ============================================================
def perform_cross_validation(X, y, cv_folds=10):
    print(f"\n PERFORMING {cv_folds}-FOLD CROSS-VALIDATION")
    print("=" * 50)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    scoring = {
        "accuracy": "accuracy",
        "precision": "precision",
        "recall": "recall",
        "f1": "f1",
        "roc_auc": "roc_auc",
        "average_precision": "average_precision"
    }

    cv_results = cross_validate(
        pipe, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,␣
 ↪random_state=RANDOM_STATE),
        scoring=scoring,
        return_train_score=True,
        n_jobs=-1
    )

    print(" CROSS-VALIDATION RESULTS (Mean ± Std):")
    print("-" * 40)

    metrics_summary = {}
```

```
    for metric in scoring.keys():
        test_scores = cv_results[f"test_{metric}"]
        train_scores = cv_results[f"train_{metric}"]

        test_mean = float(np.mean(test_scores))
        test_std = float(np.std(test_scores))
        train_mean = float(np.mean(train_scores))

        metrics_summary[metric] = {
            "test_mean": test_mean,
            "test_std": test_std,
            "train_mean": train_mean
        }

        print(f"{metric.upper():<15}: {test_mean:.4f} ± {test_std:.4f} (train:␣
↪{train_mean:.4f})")

    overfit_gap = metrics_summary["accuracy"]["train_mean"] -␣
↪metrics_summary["accuracy"]["test_mean"]
    print(f"\n Overfitting Analysis:")
    print(f"   Train-Test Accuracy Gap: {overfit_gap:.4f}")
    print("     Potential overfitting detected" if overfit_gap > 0.05 else "  ␣
↪ Model generalizes well")

    return metrics_summary, cv_results


# ============================================================
# 5) TRAIN/TEST/VALIDATION SPLITS + TRAINING
# ============================================================
def create_validation_set(X, y, validation_size=0.2):
    print(f"\n CREATING VALIDATION SET ({validation_size*100:.0f}% of data)")
    print("=" * 50)

    X_train_test, X_val, y_train_test, y_val = train_test_split(
        X, y, test_size=validation_size, stratify=y, random_state=RANDOM_STATE
    )

    X_train, X_test, y_train, y_test = train_test_split(
        X_train_test, y_train_test, test_size=0.25, stratify=y_train_test,␣
↪random_state=RANDOM_STATE
    )

    print(f" Training set: {X_train.shape[0]} samples")
    print(f" Test set: {X_test.shape[0]} samples")
    print(f" Validation set: {X_val.shape[0]} samples")
```

```
    return X_train, X_test, X_val, y_train, y_test, y_val


def train_with_comprehensive_validation(X, y, use_validation_set=True):
    print("\n MODEL TRAINING: Random Forest")
    print("-" * 40)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    if use_validation_set:
        X_train, X_test, X_val, y_train, y_test, y_val =␣
 ↪create_validation_set(X, y)
        print(" Training model with regularization...")
        pipe.fit(X_train, y_train)
        return pipe, X_train, X_test, X_val, y_train, y_test, y_val

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, stratify=y, random_state=RANDOM_STATE
    )
    print(f" Training set: {X_train.shape[0]} samples")
    print(f" Test set: {X_test.shape[0]} samples")
    print(" Training model with regularization...")
    pipe.fit(X_train, y_train)
    return pipe, X_train, X_test, None, y_train, y_test, None


# ============================================================
# 6) EVALUATION
# ============================================================
def evaluate_model_comprehensive(model, X_test, y_test, dataset_name="Test"):
    y_pred = model.predict(X_test)
```

```
    y_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    print("\n" + "=" * 60)
    print(f"  {dataset_name.upper()} SET COMPREHENSIVE EVALUATION")
    print("=" * 60)
    print(f"  Accuracy:  {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall:    {rec:.4f}")
    print(f"  F1-Score:  {f1:.4f}")
    print(f"  ROC-AUC:   {roc_auc:.4f}")
    print(f"  PR-AUC:    {pr_auc:.4f}")

    print(f"\n  {dataset_name} Classification Report:")
    print(classification_report(y_test, y_pred, digits=3))

    cm = confusion_matrix(y_test, y_pred)
    print(f"  {dataset_name} Confusion Matrix:")
    print("[[TN FP]\n [FN TP]]")
    print(cm)

    return y_pred, y_proba, {
        "accuracy": acc, "precision": prec, "recall": rec, "f1": f1,
        "roc_auc": roc_auc, "pr_auc": pr_auc
    }


# ============================================================
# 7) VISUALIZATIONS (UPDATED COLORS ADDED)
# ============================================================
def create_validation_comparison_plot(cv_metrics, output_dir, prefix):
    metrics_names = list(cv_metrics.keys())
    test_means = [cv_metrics[m]["test_mean"] for m in metrics_names]
    test_stds = [cv_metrics[m]["test_std"] for m in metrics_names]

    plt.figure(figsize=(12, 6))
    x_pos = np.arange(len(metrics_names))

    # ---- UPDATED: bars have different colors automatically ----
    bars = plt.bar(
        x_pos,
```

```
        test_means,
        yerr=test_stds,
        capsize=5,
        alpha=0.7
    )
    for i, b in enumerate(bars):
        b.set_color(f"C{i}")
    # ------------------------------------------------------------

    plt.xlabel("Performance Metrics")
    plt.ylabel("Score")
    plt.title(f"{len(metrics_names)} Metrics Cross-Validation Performance␣
↪({prefix.upper()})")
    plt.xticks(x_pos, [m.upper() for m in metrics_names], rotation=0)
    plt.ylim(0, 1)
    plt.grid(True, alpha=0.3, axis="y")

    for i, (mean, std) in enumerate(zip(test_means, test_stds)):
        plt.text(i, mean + 0.02, f"{mean:.3f}±{std:.3f}", ha="center",␣
↪va="bottom", fontsize=10)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_cross_validation_results.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()


def create_comprehensive_visualizations(y_test, y_proba, model, feature_names,␣
↪output_dir, prefix, cv_metrics=None):
    os.makedirs(output_dir, exist_ok=True)

    rf_model = model.named_steps["rf"]
    imp_df = pd.DataFrame({
        "feature": feature_names,
        "importance": rf_model.feature_importances_
    }).sort_values("importance", ascending=False)

    imp_df.to_csv(f"{output_dir}/{prefix}_feature_importances.csv", index=False)

    # Feature importance plot
    plt.figure(figsize=(12, 8))
    top_features = imp_df.head(15).iloc[::-1]
    plt.barh(range(len(top_features)), top_features["importance"], alpha=0.7)
    plt.yticks(range(len(top_features)), top_features["feature"])
    plt.xlabel("Feature Importance Score")
    plt.title(f"Top 15 Features for {prefix.upper()} Prediction")
    plt.grid(True, alpha=0.3, axis="x")
```

```
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_feature_importances.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # ROC + PR curves
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    roc_auc = roc_auc_score(y_test, y_proba)

    # ---- UPDATED: ROC curve line color ----
    ax1.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=2, color="C0")
    # ---------------------------------------

    ax1.plot([0, 1], [0, 1], "k--", alpha=0.5)
    ax1.set_xlabel("False Positive Rate")
    ax1.set_ylabel("True Positive Rate")
    ax1.set_title(f"ROC Curve ({prefix.upper()})")
    ax1.legend()
    ax1.grid(True, alpha=0.3)

    precision_vals, recall_vals, _ = precision_recall_curve(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    # ---- UPDATED: PR curve line color ----
    ax2.plot(recall_vals, precision_vals, label=f"AP = {pr_auc:.3f}",␣
↪linewidth=2, color="C1")
    # --------------------------------------

    ax2.set_xlabel("Recall")
    ax2.set_ylabel("Precision")
    ax2.set_title(f"Precision-Recall Curve ({prefix.upper()})")
    ax2.legend()
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_roc_pr_curves.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # Confusion matrix heatmap (threshold 0.5)
    y_pred = (y_proba >= 0.5).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
```

```
                xticklabels=["Pred 0", "Pred 1"],
                yticklabels=["True 0", "True 1"])
    plt.title(f"Confusion Matrix ({prefix.upper()})")
    plt.ylabel("True Label")
    plt.xlabel("Predicted Label")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_confusion_matrix.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()

    if cv_metrics:
        create_validation_comparison_plot(cv_metrics, output_dir, prefix)

    return imp_df


# ============================================================
# 8) SAVE RESULTS (GENERALIZED)
# ============================================================
def save_comprehensive_results(model, metrics, imp_df, df_clean, target_col,␣
 ↪feature_names, output_dir, prefix, cv_metrics=None):
    print(f"\n SAVING OUTPUTS to {output_dir}/ ...")
    os.makedirs(output_dir, exist_ok=True)

    # Metrics text file
    with open(f"{output_dir}/{prefix}_comprehensive_metrics.txt", "w",␣
 ↪encoding="utf-8") as f:
        f.write(f"COMPREHENSIVE RANDOM FOREST RESULTS ({prefix.upper()})\n")
        f.write("=" * 60 + "\n")
        f.write(f"Dataset shape: {df_clean.shape}\n")
        f.write(f"Target column: {target_col}\n")
        f.write(f"Features used: {len(feature_names)}\n\n")

        f.write("TEST SET METRICS:\n")
        for k, v in metrics.items():
            f.write(f"{k}: {v:.4f}\n")
        f.write("\n")

        if cv_metrics:
            f.write("CROSS-VALIDATION (Mean ± Std):\n")
            for metric, scores in cv_metrics.items():
                f.write(f"{metric}: {scores['test_mean']:.4f} ±␣
 ↪{scores['test_std']:.4f}\n")
            f.write("\n")

        f.write("TOP 10 FEATURES:\n")
        for _, row in imp_df.head(10).iterrows():
```

```
            f.write(f"{row['feature']}: {row['importance']:.6f}\n")

    # Predictions on full dataset (IMPORTANT: use the same feature columns)
    X_full = df_clean.drop(columns=[target_col], errors="ignore")
    X_full = X_full.reindex(columns=feature_names)  # align to training feature␣
 ↪set

    preds = model.predict_proba(X_full)[:, 1]
    predictions_df = df_clean.copy()
    predictions_df[f"{prefix}_Pred_Prob"] = preds
    predictions_df[f"{prefix}_Pred"] = (preds >= 0.5).astype(int)
    predictions_df[f"{prefix}_Correct"] = (predictions_df[f"{prefix}_Pred"] ==␣
 ↪predictions_df[target_col]).astype(int)

    predictions_df.to_excel(f"{output_dir}/{prefix}_predictions.xlsx",␣
 ↪index=False)

    # Save model
    joblib.dump(model, f"{output_dir}/{prefix}_rf_model.joblib")

    print("  Saved metrics, plots, predictions, and model.")


# ============================================================
# 9) MAIN PIPELINE (ONE FUNCTION FOR ANY BFx)
# ============================================================
def run_comprehensive_bf_analysis(
    target_bf="BF1",
    use_cross_validation=True,
    use_validation_set=True,
    df=None
):
    """
    Run the full pipeline for a given target_bf (e.g., BF1, BF3, BF4, BF5).
    If df is None, it will prompt you to upload/load.
    """
    try:
        prefix = str(target_bf).strip().lower().replace(" ", "").replace("_",␣
 ↪"")
        output_dir = f"{prefix}_comprehensive_outputs"

        print("  COMPREHENSIVE MICROFINANCE ANALYSIS")
        print("=" * 60)
        print(f"  Target: {target_bf}")
        print(f"  Output folder: {output_dir}/")
        print("=" * 60)
```

```
        # Load data if df not provided
        if df is None:
            try:
                import google.colab  # noqa: F401
                df = load_excel_from_colab()
            except Exception:
                df = load_excel_locally()

        print(f"  Initial dataset shape: {df.shape}")

        # Detect + preprocess target
        target_col = detect_bf_target(df, target_bf=target_bf)
        y, valid_mask = preprocess_bf_target(df[target_col],␣
↪target_label=target_col)
        df_clean = df.loc[valid_mask].copy()
        print(f"  Clean dataset: {df_clean.shape} (after removing invalid␣
↪target rows)")

        # Features
        X_final, feature_names = build_feature_matrix(df_clean, target_col)

        # CV (10-fold by default) with 6 metrics
        cv_metrics = None
        if use_cross_validation:
            cv_metrics, _ = perform_cross_validation(X_final, y, cv_folds=10)

        # Train
        model, X_train, X_test, X_val, y_train, y_test, y_val =␣
↪train_with_comprehensive_validation(
            X_final, y, use_validation_set=use_validation_set
        )

        # Evaluate test
        _, y_proba, test_metrics = evaluate_model_comprehensive(model, X_test,␣
↪y_test, "Test")

        # (Optional) Evaluate val
        if X_val is not None and y_val is not None:
            evaluate_model_comprehensive(model, X_val, y_val, "Validation")

        # Visuals
        imp_df = create_comprehensive_visualizations(
            y_test=y_test,
            y_proba=y_proba,
            model=model,
            feature_names=feature_names,
            output_dir=output_dir,
```

```
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        # Save
        save_comprehensive_results(
            model=model,
            metrics=test_metrics,
            imp_df=imp_df,
            df_clean=df_clean,
            target_col=target_col,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        print("\n DONE.")
        return model, test_metrics, imp_df, cv_metrics

    except Exception as e:
        print(f"\n Analysis failed: {type(e).__name__}: {e}")
        import traceback
        traceback.print_exc(limit=2)
        return None, None, None, None


# ========= USAGE EXAMPLES =========
if __name__ == "__main__":
    # Single target run:
    run_comprehensive_bf_analysis(target_bf="Bf4", use_cross_validation=True,␣
 ↪use_validation_set=True)

    # Multiple targets (run one by one):
    # for bf in ["BF1", "BF3", "BF4", "BF5"]:
    #     run_comprehensive_bf_analysis(target_bf=bf,␣
 ↪use_cross_validation=True, use_validation_set=True)
```

```
 COMPREHENSIVE MICROFINANCE ANALYSIS
============================================================
 Target: Bf4
 Output folder: bf4_comprehensive_outputs/
============================================================
 Please select the Excel file with your encoded dataset…
```

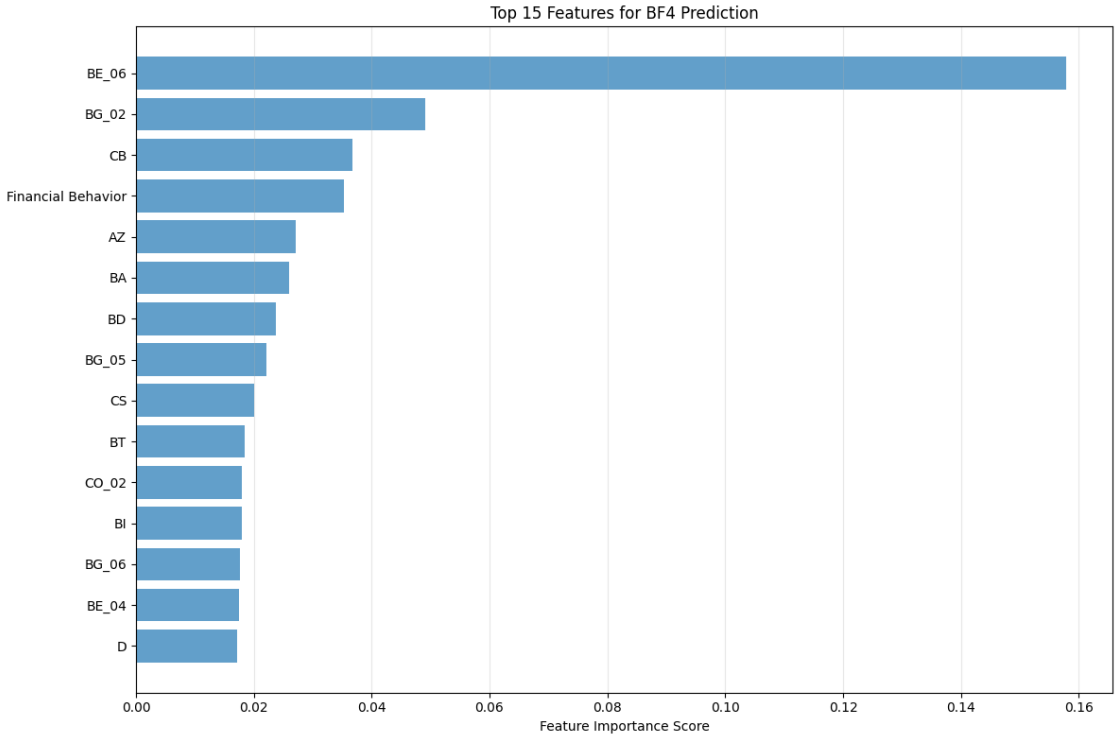
```
<IPython.core.display.HTML object>
```

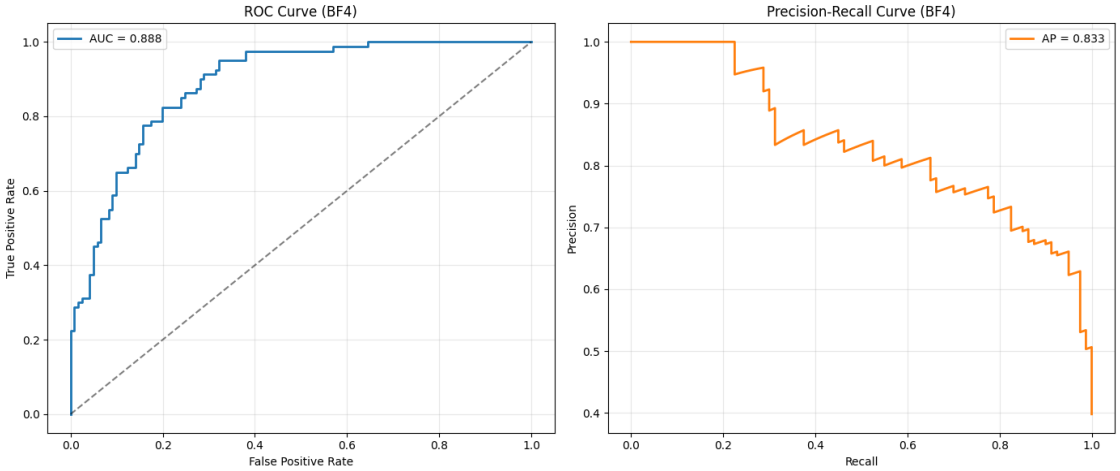
```
Saving rf_dataset_encoded_dataset.xlsx to rf_dataset_encoded_dataset (2).xlsx
```

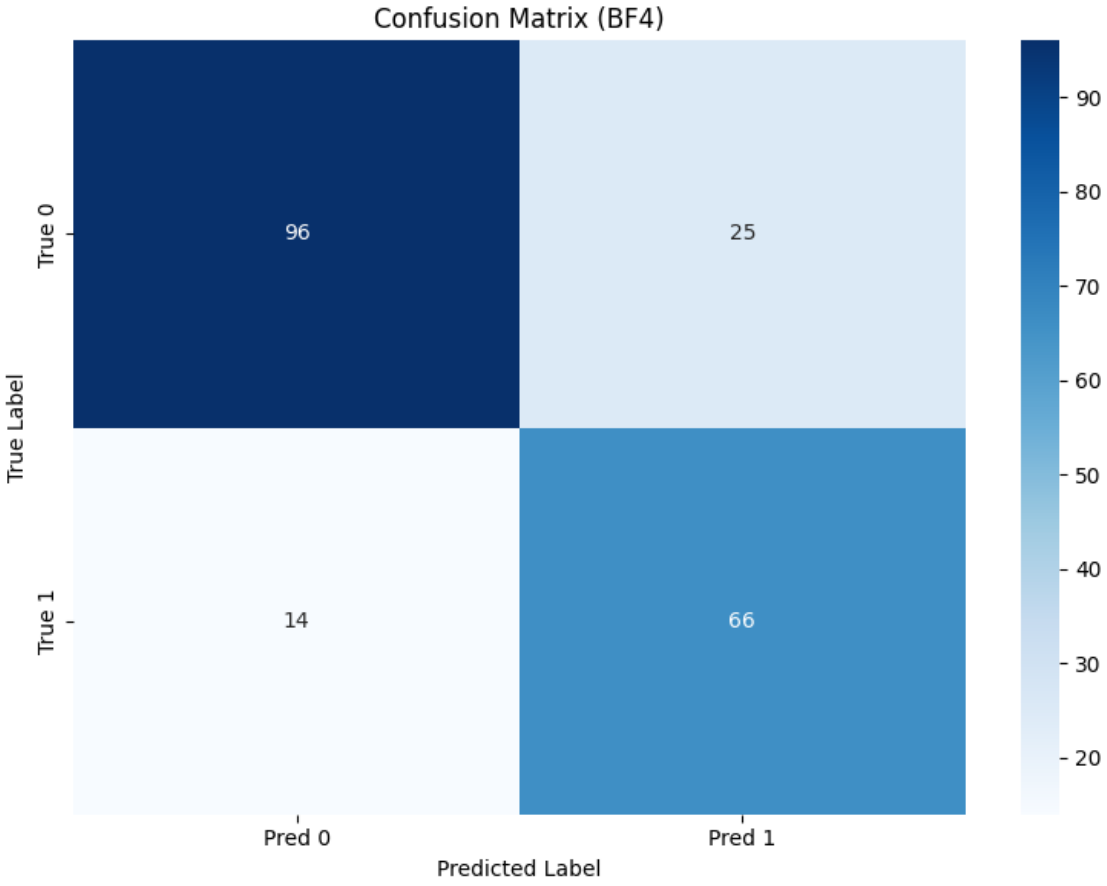
```
 Loaded dataset: (1002, 85) (rows, cols)
 Initial dataset shape: (1002, 85)
 Bf4 column not found automatically.
 Available columns:
['D', 'AY', 'AZ', 'BA', 'BB', 'BC', 'BD', 'BE_01', 'BE_02', 'BE_03', 'BE_04',
'BE_05', 'BE_06', 'BE_07', 'BE_08', 'BE_09', 'BE_10', 'BF', 'BG_01', 'BG_02',
'BG_03', 'BG_04', 'BG_05', 'BG_06', 'BH', 'BI', 'BN', 'BO', 'BP', 'BQ', 'BR',
'BS', 'BT', 'BU', 'BV', 'BW_01', 'BW_02', 'BW_03', 'BW_04', 'BW_05', 'BX', 'BY',
'CB', 'CD', 'CE', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM_01', 'CM_02', 'CM_03',
'CM_04', 'CM_05', 'CM_06', 'CN', 'CO_01', 'CO_02', 'CO_03', 'CO_04', 'CO_05',
'CO_06', 'CO_07', 'CO_08', 'CO_09', 'CP', 'CR_01', 'CR_02', 'CR_03', 'CR_04',
'CS', 'CT_01', 'CT_02', 'CT_03', 'CT_04', 'CT_05', 'CT_06', 'CU', 'Occupation',
'Financial Behavior', 'Family', 'Lifestyle', 'Digital Literacy']
 Enter exact column name for Bf4: BG_04
 Using specified target column: BG_04

 TARGET ANALYSIS: BG_04
----------------------------------------
Value distribution:
BG_04
0    605
1    397
Name: count, dtype: int64
Unique values: [np.int64(0), np.int64(1)]
 Final class balance: {0: 0.604, 1: 0.396}
 Clean dataset: (1002, 85) (after removing invalid target rows)

 FEATURE ENGINEERING
----------------------------------------
 Found 84 numeric features
 Handling missing values with median imputation…
 Final feature matrix: (1002, 84) (rows, cols)

 PERFORMING 10-FOLD CROSS-VALIDATION
================================================
 CROSS-VALIDATION RESULTS (Mean ± Std):
----------------------------------------
ACCURACY       : 0.8264 ± 0.0280 (train: 0.9155)
PRECISION      : 0.7464 ± 0.0365 (train: 0.8477)
RECALL         : 0.8539 ± 0.0459 (train: 0.9591)
F1             : 0.7958 ± 0.0328 (train: 0.9000)
ROC_AUC        : 0.9169 ± 0.0392 (train: 0.9761)
AVERAGE_PRECISION: 0.8714 ± 0.0694 (train: 0.9608)

 Overfitting Analysis:
  Train-Test Accuracy Gap: 0.0891
    Potential overfitting detected
```

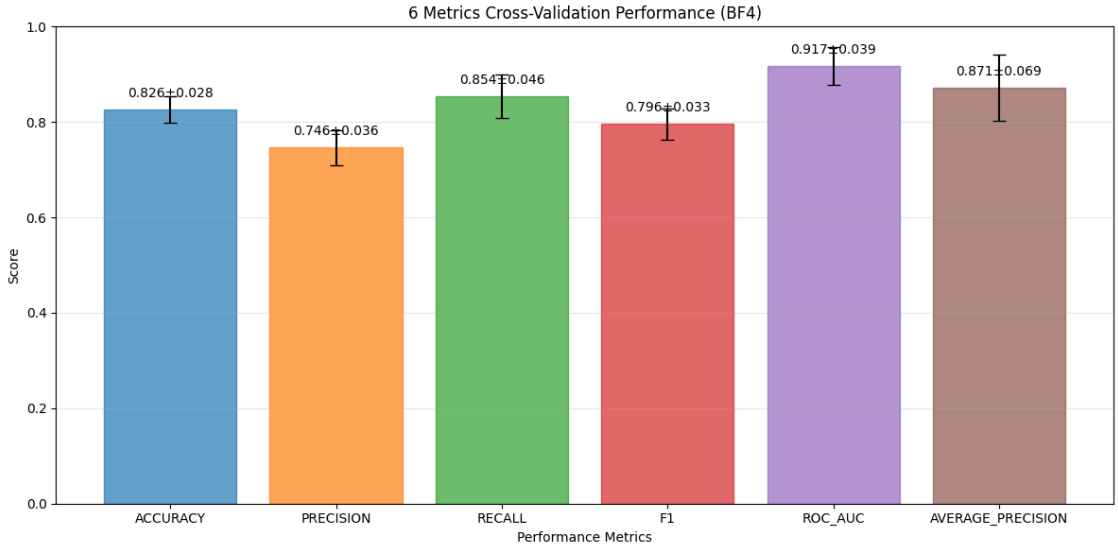
```
  MODEL TRAINING: Random Forest
--------------------------------------

  CREATING VALIDATION SET (20% of data)
==================================================
  Training set: 600 samples
  Test set: 201 samples
  Validation set: 201 samples
  Training model with regularization…

============================================================
  TEST SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.8060
  Precision: 0.7253
  Recall:    0.8250
  F1-Score:  0.7719
  ROC-AUC:   0.8883
  PR-AUC:    0.8334

  Test Classification Report:
              precision    recall  f1-score   support

           0      0.873     0.793     0.831       121
           1      0.725     0.825     0.772        80

    accuracy                          0.806       201
   macro avg      0.799     0.809     0.802       201
weighted avg      0.814     0.806     0.808       201

  Test Confusion Matrix:
[[TN FP]
 [FN TP]]
[[96 25]
 [14 66]]

============================================================
  VALIDATION SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.8259
  Precision: 0.7528
  Recall:    0.8375
  F1-Score:  0.7929
  ROC-AUC:   0.9028
  PR-AUC:    0.8450

  Validation Classification Report:
              precision    recall  f1-score   support
```

```
              0       0.884     0.818     0.850       121
              1       0.753     0.838     0.793        80

    accuracy                            0.826       201
   macro avg        0.818     0.828     0.821       201
weighted avg        0.832     0.826     0.827       201

 Validation Confusion Matrix:
[[TN FP]
 [FN TP]]
[[99 22]
 [13 67]]
```




ROC Curve (BF4)
AUC = 0.888
True Positive Rate
False Positive Rate
Precision-Recall Curve (BF4)
AP = 0.833
Precision
Recall


Confusion Matrix (BF4)
True 0
True 1
True Label
96
25
14
66
Pred 0
Pred 1
Predicted Label




```
SAVING OUTPUTS to bf4_comprehensive_outputs/ …
Saved metrics, plots, predictions, and model.

DONE.
```

[4]:
```
"""
Reusable Random Forest Pipeline for Microfinance Impact Analysis
----------------------------------------------------------------
Use this to predict BF1 / BF3 / BF4 / BF5 ... by changing ONE variable.

Example:
    run_comprehensive_bf_analysis(target_bf="BF1")
    run_comprehensive_bf_analysis(target_bf="BF3")
"""

# ===========
# 0) IMPORTS
# ===========
import os, io, re, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪cross_validate
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    accuracy_score, precision_score, recall_score, f1_score
)
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
import joblib

RANDOM_STATE = 42


# ============================================================
# 1) DATA LOADING
# ============================================================
def load_excel_from_colab():
    """Upload Excel in Google Colab."""
    try:
        from google.colab import files
        print(" Please select the Excel file with your encoded dataset...")
        uploaded = files.upload()
        path = list(uploaded.keys())[0]
        df = pd.read_excel(path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        raise RuntimeError("Colab upload failed.") from e


def load_excel_locally():
    """Upload Excel when running locally."""
    try:
        import tkinter as tk
        from tkinter import filedialog
        print(" Please select the Excel file...")
        root = tk.Tk()
        root.withdraw()
        file_path = filedialog.askopenfilename(
            title="Select Excel file",
            filetypes=[("Excel files", "*.xlsx *.xls")]
        )
        if not file_path:
            raise ValueError("No file selected")
        df = pd.read_excel(file_path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
```

```
        return df
    except Exception as e:
        print(f" File dialog failed: {e}")
        file_path = input(" Enter full path to your Excel file: ")
        df = pd.read_excel(file_path)
        print(f" Loaded dataset: {df.shape} (rows, cols)")
        return df


# ============================================================
# 2) TARGET SELECTION (GENERALIZED BFx)
# ============================================================
def detect_bf_target(df: pd.DataFrame, target_bf: str = "BF1") -> str:
    """
    Detect BF target column robustly.
    target_bf examples: "BF1", "BF3", "BF5"
    """
    m = re.match(r"^\s*BF\s*(\d+)\s*$", str(target_bf), flags=re.I)
    if not m:
        raise ValueError(f"target_bf must look like 'BF1', 'BF3', etc. Got:␣
↪{target_bf}")

    k = m.group(1)  # number as string
    aliases = [
        f"BF{k}", f"bf{k}",
        f"BF {k}", f"bf {k}",
        f"BF_{k}", f"bf_{k}",
        f"BF-{k}", f"bf-{k}",
    ]

    for a in aliases:
        if a in df.columns:
            print(f" Found target column: {a}")
            return a

    # case-insensitive regex search in column names
    pat = re.compile(rf"\bbf\s*[_-]?\s*{k}\b", flags=re.I)
    for col in df.columns:
        if pat.search(str(col)):
            print(f" Found target column (regex match): {col}")
            return col

    print(f" {target_bf} column not found automatically.")
    print(" Available columns:")
    print(df.columns.tolist())
    user_col = input(f" Enter exact column name for {target_bf}: ").strip()
    if user_col in df.columns:
```

```
        print(f"  Using specified target column: {user_col}")
        return user_col
    raise ValueError(f"Column '{user_col}' not found in dataset")


def preprocess_bf_target(y_series: pd.Series, target_label: str):
    """
    Convert BF target responses into binary classification (expects 0/1 after␣
 ↪cleaning).
    """
    y_numeric = pd.to_numeric(y_series, errors="coerce")
    valid_mask = y_numeric.notna() & np.isfinite(y_numeric)
    y_clean = y_numeric[valid_mask]

    if len(y_clean) == 0:
        raise ValueError(f"  No valid values for {target_label} after cleaning")

    print(f"\n  TARGET ANALYSIS: {target_label}")
    print("-" * 40)
    print("Value distribution:")
    print(y_clean.value_counts().sort_index())

    unique_vals = sorted(y_clean.unique())
    print(f"Unique values: {unique_vals}")

    # If your encoded dataset is already 0/1, this is perfect.
    # If it includes 1/2 or 1/3 scales, you should define a threshold rule here.
    y_binary = y_clean.astype(int)

    class_balance = y_binary.value_counts(normalize=True).round(3)
    print(f"  Final class balance: {class_balance.to_dict()}")

    return y_binary, valid_mask


# ============================================================
# 3) FEATURE ENGINEERING
# ============================================================
def build_feature_matrix(df_clean: pd.DataFrame, target_col: str):
    """Prepare features for ML (numeric-only)."""
    print("\n  FEATURE ENGINEERING")
    print("-" * 40)

    X = df_clean.drop(columns=[target_col], errors="ignore")
    numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
    print(f"  Found {len(numeric_cols)} numeric features")
```

```
    if len(numeric_cols) == 0:
        raise ValueError("  No numeric features found for modeling")

    print("  Handling missing values with median imputation...")
    imputer = SimpleImputer(strategy="median")
    X_imputed = imputer.fit_transform(X[numeric_cols])
    X_imputed = pd.DataFrame(X_imputed, columns=numeric_cols, index=X.index)

    non_constant_cols = [c for c in X_imputed.columns if X_imputed[c].nunique()␣
 ↪> 1]
    dropped = len(numeric_cols) - len(non_constant_cols)
    if dropped > 0:
        print(f"   Removed {dropped} constant columns")

    X_final = X_imputed[non_constant_cols]
    print(f"  Final feature matrix: {X_final.shape} (rows, cols)")
    return X_final, non_constant_cols


# ============================================================
# 4) CROSS-VALIDATION
# ============================================================
def perform_cross_validation(X, y, cv_folds=10):
    print(f"\n  PERFORMING {cv_folds}-FOLD CROSS-VALIDATION")
    print("=" * 50)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    scoring = {
        "accuracy": "accuracy",
        "precision": "precision",
```

```
        "recall": "recall",
        "f1": "f1",
        "roc_auc": "roc_auc",
        "average_precision": "average_precision"
    }

    cv_results = cross_validate(
        pipe, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,␣
↪random_state=RANDOM_STATE),
        scoring=scoring,
        return_train_score=True,
        n_jobs=-1
    )

    print(" CROSS-VALIDATION RESULTS (Mean ± Std):")
    print("-" * 40)

    metrics_summary = {}
    for metric in scoring.keys():
        test_scores = cv_results[f"test_{metric}"]
        train_scores = cv_results[f"train_{metric}"]

        test_mean = float(np.mean(test_scores))
        test_std = float(np.std(test_scores))
        train_mean = float(np.mean(train_scores))

        metrics_summary[metric] = {
            "test_mean": test_mean,
            "test_std": test_std,
            "train_mean": train_mean
        }

        print(f"{metric.upper():<15}: {test_mean:.4f} ± {test_std:.4f} (train:␣
↪{train_mean:.4f})")

    overfit_gap = metrics_summary["accuracy"]["train_mean"] -␣
↪metrics_summary["accuracy"]["test_mean"]
    print(f"\n Overfitting Analysis:")
    print(f"   Train-Test Accuracy Gap: {overfit_gap:.4f}")
    print("     Potential overfitting detected" if overfit_gap > 0.05 else "  ␣
↪ Model generalizes well")

    return metrics_summary, cv_results


# ============================================================
```

```
# 5) TRAIN/TEST/VALIDATION SPLITS + TRAINING
# ============================================================
def create_validation_set(X, y, validation_size=0.2):
    print(f"\n CREATING VALIDATION SET ({validation_size*100:.0f}% of data)")
    print("=" * 50)

    X_train_test, X_val, y_train_test, y_val = train_test_split(
        X, y, test_size=validation_size, stratify=y, random_state=RANDOM_STATE
    )

    X_train, X_test, y_train, y_test = train_test_split(
        X_train_test, y_train_test, test_size=0.25, stratify=y_train_test,␣
 ↪random_state=RANDOM_STATE
    )

    print(f" Training set: {X_train.shape[0]} samples")
    print(f" Test set: {X_test.shape[0]} samples")
    print(f" Validation set: {X_val.shape[0]} samples")

    return X_train, X_test, X_val, y_train, y_test, y_val


def train_with_comprehensive_validation(X, y, use_validation_set=True):
    print("\n MODEL TRAINING: Random Forest")
    print("-" * 40)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    if use_validation_set:
        X_train, X_test, X_val, y_train, y_test, y_val =␣
 ↪create_validation_set(X, y)
```

```
        print("  Training model with regularization...")
        pipe.fit(X_train, y_train)
        return pipe, X_train, X_test, X_val, y_train, y_test, y_val

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, stratify=y, random_state=RANDOM_STATE
    )
    print(f"  Training set: {X_train.shape[0]} samples")
    print(f"  Test set: {X_test.shape[0]} samples")
    print("  Training model with regularization...")
    pipe.fit(X_train, y_train)
    return pipe, X_train, X_test, None, y_train, y_test, None


# ============================================================
# 6) EVALUATION
# ============================================================
def evaluate_model_comprehensive(model, X_test, y_test, dataset_name="Test"):
    y_pred = model.predict(X_test)
    y_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    print("\n" + "=" * 60)
    print(f"  {dataset_name.upper()} SET COMPREHENSIVE EVALUATION")
    print("=" * 60)
    print(f"  Accuracy:  {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall:    {rec:.4f}")
    print(f"  F1-Score:  {f1:.4f}")
    print(f"  ROC-AUC:   {roc_auc:.4f}")
    print(f"  PR-AUC:    {pr_auc:.4f}")

    print(f"\n {dataset_name} Classification Report:")
    print(classification_report(y_test, y_pred, digits=3))

    cm = confusion_matrix(y_test, y_pred)
    print(f"  {dataset_name} Confusion Matrix:")
    print("[[TN FP]\n [FN TP]]")
    print(cm)

    return y_pred, y_proba, {
```

```
        "accuracy": acc, "precision": prec, "recall": rec, "f1": f1,
        "roc_auc": roc_auc, "pr_auc": pr_auc
    }


# ============================================================
# 7) VISUALIZATIONS (UPDATED COLORS ADDED)
# ============================================================
def create_validation_comparison_plot(cv_metrics, output_dir, prefix):
    metrics_names = list(cv_metrics.keys())
    test_means = [cv_metrics[m]["test_mean"] for m in metrics_names]
    test_stds = [cv_metrics[m]["test_std"] for m in metrics_names]

    plt.figure(figsize=(12, 6))
    x_pos = np.arange(len(metrics_names))

    # ---- UPDATED: bars have different colors automatically ----
    bars = plt.bar(
        x_pos,
        test_means,
        yerr=test_stds,
        capsize=5,
        alpha=0.7
    )
    for i, b in enumerate(bars):
        b.set_color(f"C{i}")
    # -----------------------------------------------------------

    plt.xlabel("Performance Metrics")
    plt.ylabel("Score")
    plt.title(f"{len(metrics_names)} Metrics Cross-Validation Performance␣
↪({prefix.upper()})")
    plt.xticks(x_pos, [m.upper() for m in metrics_names], rotation=0)
    plt.ylim(0, 1)
    plt.grid(True, alpha=0.3, axis="y")

    for i, (mean, std) in enumerate(zip(test_means, test_stds)):
        plt.text(i, mean + 0.02, f"{mean:.3f}±{std:.3f}", ha="center",␣
↪va="bottom", fontsize=10)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_cross_validation_results.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()
```

```
def create_comprehensive_visualizations(y_test, y_proba, model, feature_names,␣
 ↪output_dir, prefix, cv_metrics=None):
    os.makedirs(output_dir, exist_ok=True)

    rf_model = model.named_steps["rf"]
    imp_df = pd.DataFrame({
        "feature": feature_names,
        "importance": rf_model.feature_importances_
    }).sort_values("importance", ascending=False)

    imp_df.to_csv(f"{output_dir}/{prefix}_feature_importances.csv", index=False)

    # Feature importance plot
    plt.figure(figsize=(12, 8))
    top_features = imp_df.head(15).iloc[::-1]
    plt.barh(range(len(top_features)), top_features["importance"], alpha=0.7)
    plt.yticks(range(len(top_features)), top_features["feature"])
    plt.xlabel("Feature Importance Score")
    plt.title(f"Top 15 Features for {prefix.upper()} Prediction")
    plt.grid(True, alpha=0.3, axis="x")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_feature_importances.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()

    # ROC + PR curves
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    roc_auc = roc_auc_score(y_test, y_proba)

    # ---- UPDATED: ROC curve line color ----
    ax1.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=2, color="C0")
    # ---------------------------------------

    ax1.plot([0, 1], [0, 1], "k--", alpha=0.5)
    ax1.set_xlabel("False Positive Rate")
    ax1.set_ylabel("True Positive Rate")
    ax1.set_title(f"ROC Curve ({prefix.upper()})")
    ax1.legend()
    ax1.grid(True, alpha=0.3)

    precision_vals, recall_vals, _ = precision_recall_curve(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    # ---- UPDATED: PR curve line color ----
```

```
    ax2.plot(recall_vals, precision_vals, label=f"AP = {pr_auc:.3f}",␣
↪linewidth=2, color="C1")
    # ------------------------------------

    ax2.set_xlabel("Recall")
    ax2.set_ylabel("Precision")
    ax2.set_title(f"Precision-Recall Curve ({prefix.upper()})")
    ax2.legend()
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_roc_pr_curves.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # Confusion matrix heatmap (threshold 0.5)
    y_pred = (y_proba >= 0.5).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
                xticklabels=["Pred 0", "Pred 1"],
                yticklabels=["True 0", "True 1"])
    plt.title(f"Confusion Matrix ({prefix.upper()})")
    plt.ylabel("True Label")
    plt.xlabel("Predicted Label")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_confusion_matrix.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    if cv_metrics:
        create_validation_comparison_plot(cv_metrics, output_dir, prefix)

    return imp_df


# ============================================================
# 8) SAVE RESULTS (GENERALIZED)
# ============================================================
def save_comprehensive_results(model, metrics, imp_df, df_clean, target_col,␣
↪feature_names, output_dir, prefix, cv_metrics=None):
    print(f"\n SAVING OUTPUTS to {output_dir}/ ...")
    os.makedirs(output_dir, exist_ok=True)

    # Metrics text file
```

```
    with open(f"{output_dir}/{prefix}_comprehensive_metrics.txt", "w",␣
↪encoding="utf-8") as f:
        f.write(f"COMPREHENSIVE RANDOM FOREST RESULTS ({prefix.upper()})\n")
        f.write("=" * 60 + "\n")
        f.write(f"Dataset shape: {df_clean.shape}\n")
        f.write(f"Target column: {target_col}\n")
        f.write(f"Features used: {len(feature_names)}\n\n")

        f.write("TEST SET METRICS:\n")
        for k, v in metrics.items():
            f.write(f"{k}: {v:.4f}\n")
        f.write("\n")

        if cv_metrics:
            f.write("CROSS-VALIDATION (Mean ± Std):\n")
            for metric, scores in cv_metrics.items():
                f.write(f"{metric}: {scores['test_mean']:.4f} ±␣
↪{scores['test_std']:.4f}\n")
            f.write("\n")

        f.write("TOP 10 FEATURES:\n")
        for _, row in imp_df.head(10).iterrows():
            f.write(f"{row['feature']}: {row['importance']:.6f}\n")

    # Predictions on full dataset (IMPORTANT: use the same feature columns)
    X_full = df_clean.drop(columns=[target_col], errors="ignore")
    X_full = X_full.reindex(columns=feature_names)  # align to training feature␣
↪set

    preds = model.predict_proba(X_full)[:, 1]
    predictions_df = df_clean.copy()
    predictions_df[f"{prefix}_Pred_Prob"] = preds
    predictions_df[f"{prefix}_Pred"] = (preds >= 0.5).astype(int)
    predictions_df[f"{prefix}_Correct"] = (predictions_df[f"{prefix}_Pred"] ==␣
↪predictions_df[target_col]).astype(int)

    predictions_df.to_excel(f"{output_dir}/{prefix}_predictions.xlsx",␣
↪index=False)

    # Save model
    joblib.dump(model, f"{output_dir}/{prefix}_rf_model.joblib")

    print("  Saved metrics, plots, predictions, and model.")


# ============================================================
# 9) MAIN PIPELINE (ONE FUNCTION FOR ANY BFx)
```

```python
# ============================================================
def run_comprehensive_bf_analysis(
    target_bf="BF1",
    use_cross_validation=True,
    use_validation_set=True,
    df=None
):
    """
    Run the full pipeline for a given target_bf (e.g., BF1, BF3, BF4, BF5).
    If df is None, it will prompt you to upload/load.
    """
    try:
        prefix = str(target_bf).strip().lower().replace(" ", "").replace("_",␣
 ↪"")
        output_dir = f"{prefix}_comprehensive_outputs"

        print(" COMPREHENSIVE MICROFINANCE ANALYSIS")
        print("=" * 60)
        print(f" Target: {target_bf}")
        print(f" Output folder: {output_dir}/")
        print("=" * 60)

        # Load data if df not provided
        if df is None:
            try:
                import google.colab  # noqa: F401
                df = load_excel_from_colab()
            except Exception:
                df = load_excel_locally()

        print(f" Initial dataset shape: {df.shape}")

        # Detect + preprocess target
        target_col = detect_bf_target(df, target_bf=target_bf)
        y, valid_mask = preprocess_bf_target(df[target_col],␣
 ↪target_label=target_col)
        df_clean = df.loc[valid_mask].copy()
        print(f" Clean dataset: {df_clean.shape} (after removing invalid␣
 ↪target rows)")

        # Features
        X_final, feature_names = build_feature_matrix(df_clean, target_col)

        # CV (10-fold by default) with 6 metrics
        cv_metrics = None
        if use_cross_validation:
            cv_metrics, _ = perform_cross_validation(X_final, y, cv_folds=10)
```

```
        # Train
        model, X_train, X_test, X_val, y_train, y_test, y_val =␣
↪train_with_comprehensive_validation(
            X_final, y, use_validation_set=use_validation_set
        )

        # Evaluate test
        _, y_proba, test_metrics = evaluate_model_comprehensive(model, X_test,␣
↪y_test, "Test")

        # (Optional) Evaluate val
        if X_val is not None and y_val is not None:
            evaluate_model_comprehensive(model, X_val, y_val, "Validation")

        # Visuals
        imp_df = create_comprehensive_visualizations(
            y_test=y_test,
            y_proba=y_proba,
            model=model,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        # Save
        save_comprehensive_results(
            model=model,
            metrics=test_metrics,
            imp_df=imp_df,
            df_clean=df_clean,
            target_col=target_col,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        print("\n DONE.")
        return model, test_metrics, imp_df, cv_metrics

    except Exception as e:
        print(f"\n Analysis failed: {type(e).__name__}: {e}")
        import traceback
        traceback.print_exc(limit=2)
        return None, None, None, None
```

```python
# ========= USAGE EXAMPLES =========
if __name__ == "__main__":
    # Single target run:
    run_comprehensive_bf_analysis(target_bf="BF5", use_cross_validation=True,␣
 ↪use_validation_set=True)

    # Multiple targets (run one by one):
    # for bf in ["BF1", "BF3", "BF4", "BF5"]:
    #     run_comprehensive_bf_analysis(target_bf=bf,␣
 ↪use_cross_validation=True, use_validation_set=True)
```

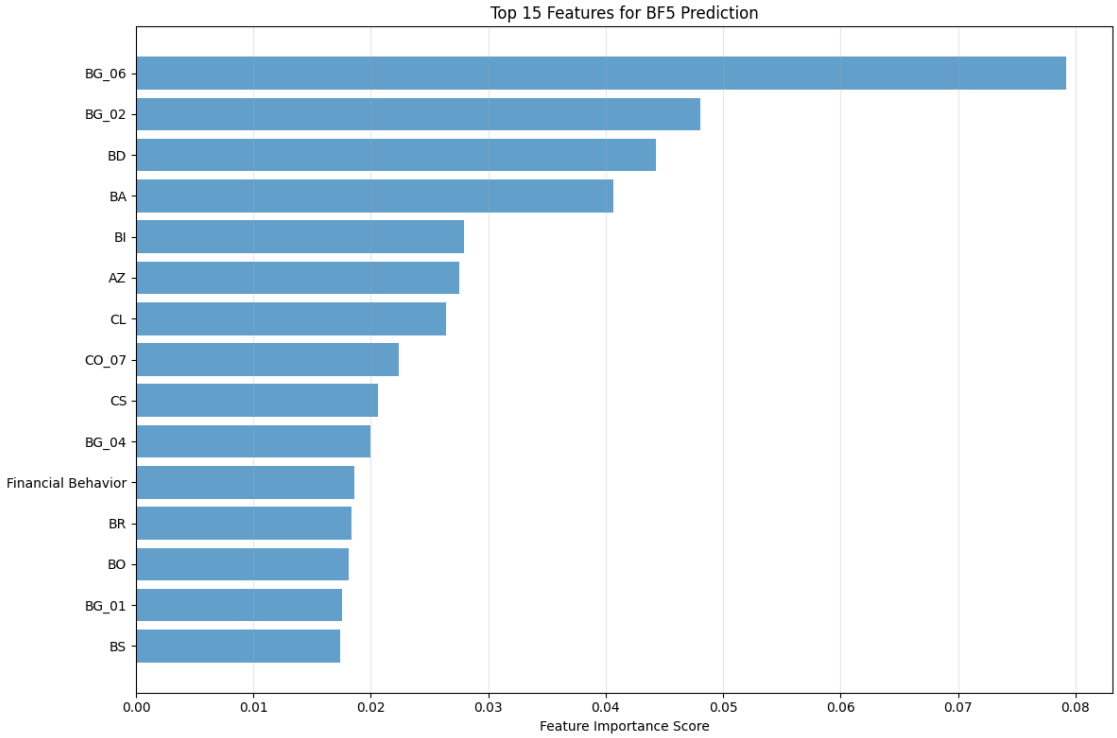
```
 COMPREHENSIVE MICROFINANCE ANALYSIS
============================================================
 Target: BF5
 Output folder: bf5_comprehensive_outputs/
============================================================
 Please select the Excel file with your encoded dataset…
```

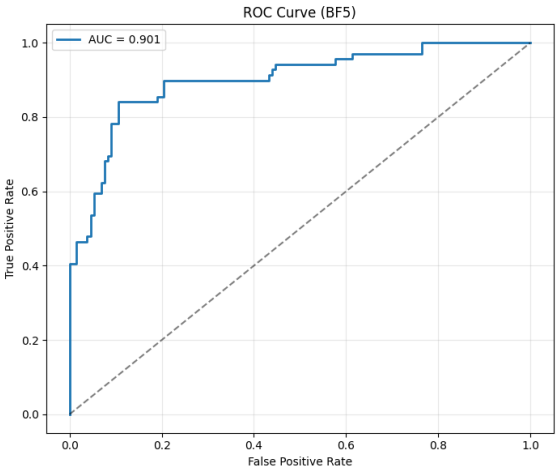
```
<IPython.core.display.HTML object>
```

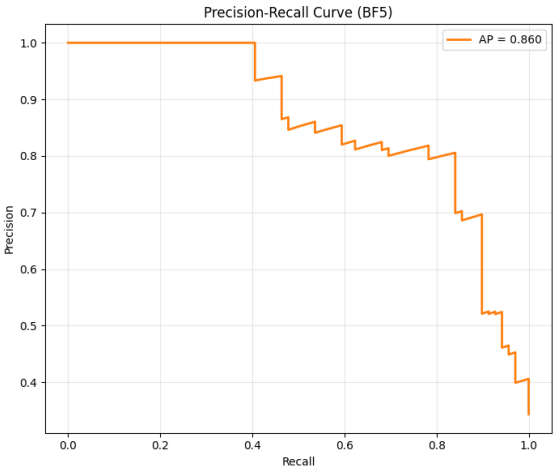
```
Saving rf_dataset_encoded_dataset.xlsx to rf_dataset_encoded_dataset (3).xlsx
 Loaded dataset: (1002, 85) (rows, cols)
 Initial dataset shape: (1002, 85)
 BF5 column not found automatically.
 Available columns:
['D', 'AY', 'AZ', 'BA', 'BB', 'BC', 'BD', 'BE_01', 'BE_02', 'BE_03', 'BE_04',
'BE_05', 'BE_06', 'BE_07', 'BE_08', 'BE_09', 'BE_10', 'BF', 'BG_01', 'BG_02',
'BG_03', 'BG_04', 'BG_05', 'BG_06', 'BH', 'BI', 'BN', 'BO', 'BP', 'BQ', 'BR',
'BS', 'BT', 'BU', 'BV', 'BW_01', 'BW_02', 'BW_03', 'BW_04', 'BW_05', 'BX', 'BY',
'CB', 'CD', 'CE', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM_01', 'CM_02', 'CM_03',
'CM_04', 'CM_05', 'CM_06', 'CN', 'CO_01', 'CO_02', 'CO_03', 'CO_04', 'CO_05',
'CO_06', 'CO_07', 'CO_08', 'CO_09', 'CP', 'CR_01', 'CR_02', 'CR_03', 'CR_04',
'CS', 'CT_01', 'CT_02', 'CT_03', 'CT_04', 'CT_05', 'CT_06', 'CU', 'Occupation',
'Financial Behavior', 'Family', 'Lifestyle', 'Digital Literacy']
 Enter exact column name for BF5: BG_05
 Using specified target column: BG_05

 TARGET ANALYSIS: BG_05
----------------------------------------
Value distribution:
BG_05
0    657
1    345
Name: count, dtype: int64
Unique values: [np.int64(0), np.int64(1)]
 Final class balance: {0: 0.656, 1: 0.344}
```

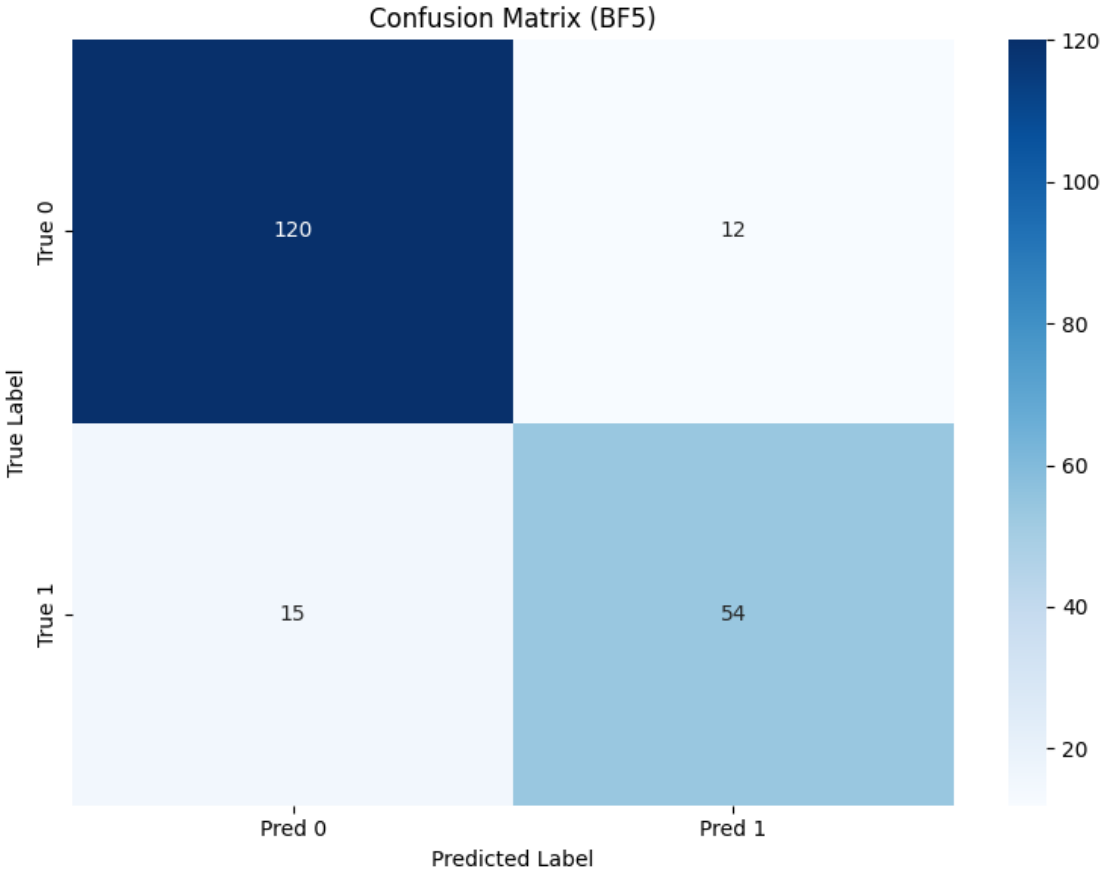
```
  Clean dataset: (1002, 85) (after removing invalid target rows)

  FEATURE ENGINEERING
----------------------------------------
  Found 84 numeric features
  Handling missing values with median imputation…
  Final feature matrix: (1002, 84) (rows, cols)

  PERFORMING 10-FOLD CROSS-VALIDATION
==================================================
  CROSS-VALIDATION RESULTS (Mean ± Std):
----------------------------------------
ACCURACY       : 0.8074 ± 0.0430 (train: 0.9495)
PRECISION      : 0.7361 ± 0.0729 (train: 0.9232)
RECALL         : 0.6865 ± 0.0659 (train: 0.9311)
F1             : 0.7101 ± 0.0676 (train: 0.9270)
ROC_AUC        : 0.8976 ± 0.0348 (train: 0.9878)
AVERAGE_PRECISION: 0.8345 ± 0.0571 (train: 0.9759)

  Overfitting Analysis:
    Train-Test Accuracy Gap: 0.1422
       Potential overfitting detected

  MODEL TRAINING: Random Forest
----------------------------------------

  CREATING VALIDATION SET (20% of data)
==================================================
  Training set: 600 samples
  Test set: 201 samples
  Validation set: 201 samples
  Training model with regularization…

============================================================
  TEST SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.8657
  Precision: 0.8182
  Recall:    0.7826
  F1-Score:  0.8000
  ROC-AUC:   0.9011
  PR-AUC:    0.8601

  Test Classification Report:
              precision    recall  f1-score   support

           0      0.889     0.909     0.899       132
           1      0.818     0.783     0.800        69
```

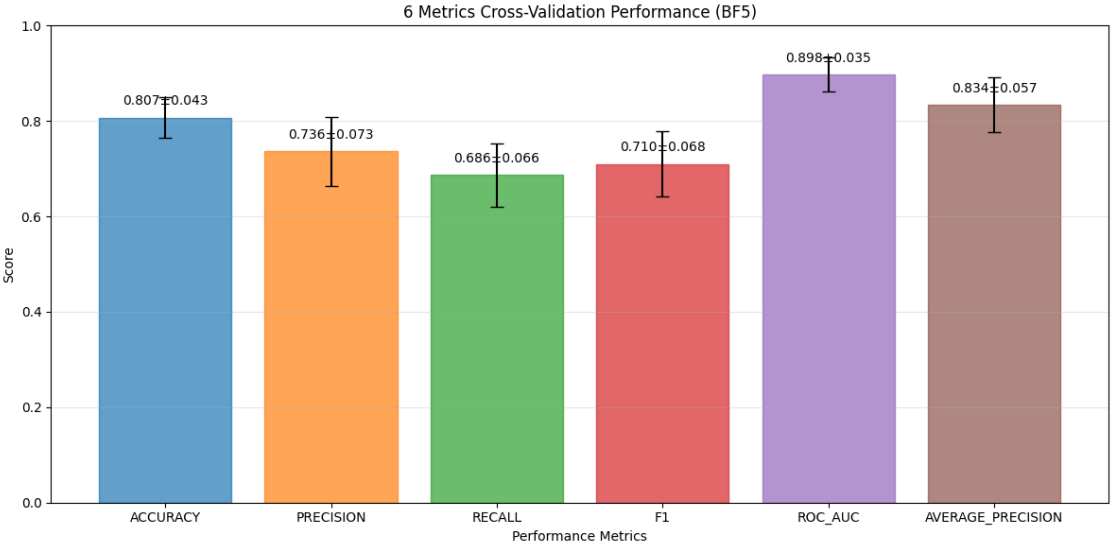
```
    accuracy                           0.866       201
   macro avg       0.854     0.846     0.849       201
weighted avg       0.865     0.866     0.865       201

 Test Confusion Matrix:
[[TN FP]
 [FN TP]]
[[120  12]
 [ 15  54]]

============================================================
 VALIDATION SET COMPREHENSIVE EVALUATION
============================================================
 Accuracy:  0.8209
 Precision: 0.7391
 Recall:    0.7391
 F1-Score:  0.7391
 ROC-AUC:   0.8813
 PR-AUC:    0.8100

 Validation Classification Report:
              precision    recall  f1-score   support

           0      0.864     0.864     0.864       132
           1      0.739     0.739     0.739        69

    accuracy                          0.821       201
   macro avg      0.801     0.801     0.801       201
weighted avg      0.821     0.821     0.821       201

 Validation Confusion Matrix:
[[TN FP]
 [FN TP]]
[[114  18]
 [ 18  51]]
```


Top 15 Features for BF5 Prediction
BG_06
BG_02
BD
BA
BI
AZ
CL
CO_07
CS
BG_04
Financial Behavior
BR
BO
BG_01
BS
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Feature Importance Score


ROC Curve (BF5)
AUC = 0.901
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0


Precision-Recall Curve (BF5)
AP = 0.860
Precision
Recall
0.4
0.5
0.6
0.7
0.8
0.9
1.0


Confusion Matrix (BF5)
True 0
True 1
True Label
120
12
15
54
Pred 0
Pred 1
Predicted Label
120
100
80
60
40
20


6 Metrics Cross-Validation Performance (BF5)
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.807±0.043
0.736±0.073
0.686±0.066
0.710±0.068
0.898±0.035
0.834±0.057
ACCURACY
PRECISION
RECALL
F1
ROC_AUC
AVERAGE_PRECISION
Performance Metrics

```
SAVING OUTPUTS to bf5_comprehensive_outputs/ …
Saved metrics, plots, predictions, and model.

DONE.
```

[5]:
```python
"""
Reusable Random Forest Pipeline for Microfinance Impact Analysis
----------------------------------------------------------------
Use this to predict BF1 / BF3 / BF4 / BF5 ... by changing ONE variable.

Example:
    run_comprehensive_bf_analysis(target_bf="BF1")
    run_comprehensive_bf_analysis(target_bf="BF3")
"""

# ===========
# 0) IMPORTS
# ===========
import os, io, re, warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold,␣
 ↪cross_validate
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_auc_score, roc_curve,
    precision_recall_curve, average_precision_score,
    accuracy_score, precision_score, recall_score, f1_score
)
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
import joblib

RANDOM_STATE = 42


# ============================================================
# 1) DATA LOADING
# ============================================================
def load_excel_from_colab():
    """Upload Excel in Google Colab."""
```

```
    try:
        from google.colab import files
        print("  Please select the Excel file with your encoded dataset...")
        uploaded = files.upload()
        path = list(uploaded.keys())[0]
        df = pd.read_excel(path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        raise RuntimeError("Colab upload failed.") from e


def load_excel_locally():
    """Upload Excel when running locally."""
    try:
        import tkinter as tk
        from tkinter import filedialog
        print("  Please select the Excel file...")
        root = tk.Tk()
        root.withdraw()
        file_path = filedialog.askopenfilename(
            title="Select Excel file",
            filetypes=[("Excel files", "*.xlsx *.xls")]
        )
        if not file_path:
            raise ValueError("No file selected")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df
    except Exception as e:
        print(f"  File dialog failed: {e}")
        file_path = input("  Enter full path to your Excel file: ")
        df = pd.read_excel(file_path)
        print(f"  Loaded dataset: {df.shape} (rows, cols)")
        return df


# ============================================================
# 2) TARGET SELECTION (GENERALIZED BFx)
# ============================================================
def detect_bf_target(df: pd.DataFrame, target_bf: str = "BF1") -> str:
    """
    Detect BF target column robustly.
    target_bf examples: "BF1", "BF3", "BF5"
    """
    m = re.match(r"^\s*BF\s*(\d+)\s*$", str(target_bf), flags=re.I)
    if not m:
```

```
        raise ValueError(f"target_bf must look like 'BF1', 'BF3', etc. Got:␣
↪{target_bf}")

    k = m.group(1)  # number as string
    aliases = [
        f"BF{k}", f"bf{k}",
        f"BF {k}", f"bf {k}",
        f"BF_{k}", f"bf_{k}",
        f"BF-{k}", f"bf-{k}",
    ]

    for a in aliases:
        if a in df.columns:
            print(f"  Found target column: {a}")
            return a

    # case-insensitive regex search in column names
    pat = re.compile(rf"\bbf\s*[_-]?\s*{k}\b", flags=re.I)
    for col in df.columns:
        if pat.search(str(col)):
            print(f"  Found target column (regex match): {col}")
            return col

    print(f"  {target_bf} column not found automatically.")
    print("  Available columns:")
    print(df.columns.tolist())
    user_col = input(f"  Enter exact column name for {target_bf}: ").strip()
    if user_col in df.columns:
        print(f"  Using specified target column: {user_col}")
        return user_col
    raise ValueError(f"Column '{user_col}' not found in dataset")


def preprocess_bf_target(y_series: pd.Series, target_label: str):
    """
    Convert BF target responses into binary classification (expects 0/1 after␣
↪cleaning).
    """
    y_numeric = pd.to_numeric(y_series, errors="coerce")
    valid_mask = y_numeric.notna() & np.isfinite(y_numeric)
    y_clean = y_numeric[valid_mask]

    if len(y_clean) == 0:
        raise ValueError(f"  No valid values for {target_label} after cleaning")

    print(f"\n  TARGET ANALYSIS: {target_label}")
    print("-" * 40)
```

```
    print("Value distribution:")
    print(y_clean.value_counts().sort_index())

    unique_vals = sorted(y_clean.unique())
    print(f"Unique values: {unique_vals}")

    # If your encoded dataset is already 0/1, this is perfect.
    # If it includes 1/2 or 1/3 scales, you should define a threshold rule here.
    y_binary = y_clean.astype(int)

    class_balance = y_binary.value_counts(normalize=True).round(3)
    print(f"  Final class balance: {class_balance.to_dict()}")

    return y_binary, valid_mask


# ============================================================
# 3) FEATURE ENGINEERING
# ============================================================
def build_feature_matrix(df_clean: pd.DataFrame, target_col: str):
    """Prepare features for ML (numeric-only)."""
    print("\n FEATURE ENGINEERING")
    print("-" * 40)

    X = df_clean.drop(columns=[target_col], errors="ignore")
    numeric_cols = X.select_dtypes(include=[np.number]).columns.tolist()
    print(f"  Found {len(numeric_cols)} numeric features")

    if len(numeric_cols) == 0:
        raise ValueError("  No numeric features found for modeling")

    print("  Handling missing values with median imputation...")
    imputer = SimpleImputer(strategy="median")
    X_imputed = imputer.fit_transform(X[numeric_cols])
    X_imputed = pd.DataFrame(X_imputed, columns=numeric_cols, index=X.index)

    non_constant_cols = [c for c in X_imputed.columns if X_imputed[c].nunique()␣
 ↪> 1]
    dropped = len(numeric_cols) - len(non_constant_cols)
    if dropped > 0:
        print(f"   Removed {dropped} constant columns")

    X_final = X_imputed[non_constant_cols]
    print(f"  Final feature matrix: {X_final.shape} (rows, cols)")
    return X_final, non_constant_cols
```

```python
# ============================================================
# 4) CROSS-VALIDATION
# ============================================================
def perform_cross_validation(X, y, cv_folds=10):
    print(f"\n PERFORMING {cv_folds}-FOLD CROSS-VALIDATION")
    print("=" * 50)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    scoring = {
        "accuracy": "accuracy",
        "precision": "precision",
        "recall": "recall",
        "f1": "f1",
        "roc_auc": "roc_auc",
        "average_precision": "average_precision"
    }

    cv_results = cross_validate(
        pipe, X, y,
        cv=StratifiedKFold(n_splits=cv_folds, shuffle=True,␣
 ↪random_state=RANDOM_STATE),
        scoring=scoring,
        return_train_score=True,
        n_jobs=-1
    )

    print(" CROSS-VALIDATION RESULTS (Mean ± Std):")
    print("-" * 40)

    metrics_summary = {}
```

```
    for metric in scoring.keys():
        test_scores = cv_results[f"test_{metric}"]
        train_scores = cv_results[f"train_{metric}"]

        test_mean = float(np.mean(test_scores))
        test_std = float(np.std(test_scores))
        train_mean = float(np.mean(train_scores))

        metrics_summary[metric] = {
            "test_mean": test_mean,
            "test_std": test_std,
            "train_mean": train_mean
        }

        print(f"{metric.upper():<15}: {test_mean:.4f} ± {test_std:.4f} (train:␣
↪{train_mean:.4f})")

    overfit_gap = metrics_summary["accuracy"]["train_mean"] -␣
↪metrics_summary["accuracy"]["test_mean"]
    print(f"\n  Overfitting Analysis:")
    print(f"   Train-Test Accuracy Gap: {overfit_gap:.4f}")
    print("     Potential overfitting detected" if overfit_gap > 0.05 else "  ␣
↪  Model generalizes well")

    return metrics_summary, cv_results


# ============================================================
# 5) TRAIN/TEST/VALIDATION SPLITS + TRAINING
# ============================================================
def create_validation_set(X, y, validation_size=0.2):
    print(f"\n  CREATING VALIDATION SET ({validation_size*100:.0f}% of data)")
    print("=" * 50)

    X_train_test, X_val, y_train_test, y_val = train_test_split(
        X, y, test_size=validation_size, stratify=y, random_state=RANDOM_STATE
    )

    X_train, X_test, y_train, y_test = train_test_split(
        X_train_test, y_train_test, test_size=0.25, stratify=y_train_test,␣
↪random_state=RANDOM_STATE
    )

    print(f"  Training set: {X_train.shape[0]} samples")
    print(f"  Test set: {X_test.shape[0]} samples")
    print(f"  Validation set: {X_val.shape[0]} samples")
```

```
    return X_train, X_test, X_val, y_train, y_test, y_val


def train_with_comprehensive_validation(X, y, use_validation_set=True):
    print("\n MODEL TRAINING: Random Forest")
    print("-" * 40)

    preprocessor = Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ])

    rf = RandomForestClassifier(
        n_estimators=100,
        max_depth=8,
        min_samples_split=10,
        min_samples_leaf=5,
        max_features="sqrt",
        class_weight="balanced_subsample",
        random_state=RANDOM_STATE,
        n_jobs=-1
    )

    pipe = Pipeline([("pre", preprocessor), ("rf", rf)])

    if use_validation_set:
        X_train, X_test, X_val, y_train, y_test, y_val =␣
 ↪create_validation_set(X, y)
        print(" Training model with regularization...")
        pipe.fit(X_train, y_train)
        return pipe, X_train, X_test, X_val, y_train, y_test, y_val

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.25, stratify=y, random_state=RANDOM_STATE
    )
    print(f" Training set: {X_train.shape[0]} samples")
    print(f" Test set: {X_test.shape[0]} samples")
    print(" Training model with regularization...")
    pipe.fit(X_train, y_train)
    return pipe, X_train, X_test, None, y_train, y_test, None


# ============================================================
# 6) EVALUATION
# ============================================================
def evaluate_model_comprehensive(model, X_test, y_test, dataset_name="Test"):
    y_pred = model.predict(X_test)
```

```
    y_proba = model.predict_proba(X_test)[:, 1]

    acc = accuracy_score(y_test, y_pred)
    prec = precision_score(y_test, y_pred, zero_division=0)
    rec = recall_score(y_test, y_pred, zero_division=0)
    f1 = f1_score(y_test, y_pred, zero_division=0)
    roc_auc = roc_auc_score(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    print("\n" + "=" * 60)
    print(f"  {dataset_name.upper()} SET COMPREHENSIVE EVALUATION")
    print("=" * 60)
    print(f"  Accuracy:  {acc:.4f}")
    print(f"  Precision: {prec:.4f}")
    print(f"  Recall:    {rec:.4f}")
    print(f"  F1-Score:  {f1:.4f}")
    print(f"  ROC-AUC:   {roc_auc:.4f}")
    print(f"  PR-AUC:    {pr_auc:.4f}")

    print(f"\n  {dataset_name} Classification Report:")
    print(classification_report(y_test, y_pred, digits=3))

    cm = confusion_matrix(y_test, y_pred)
    print(f"  {dataset_name} Confusion Matrix:")
    print("[[TN FP]\n [FN TP]]")
    print(cm)

    return y_pred, y_proba, {
        "accuracy": acc, "precision": prec, "recall": rec, "f1": f1,
        "roc_auc": roc_auc, "pr_auc": pr_auc
    }


# ============================================================
# 7) VISUALIZATIONS (UPDATED COLORS ADDED)
# ============================================================
def create_validation_comparison_plot(cv_metrics, output_dir, prefix):
    metrics_names = list(cv_metrics.keys())
    test_means = [cv_metrics[m]["test_mean"] for m in metrics_names]
    test_stds = [cv_metrics[m]["test_std"] for m in metrics_names]

    plt.figure(figsize=(12, 6))
    x_pos = np.arange(len(metrics_names))

    # ---- UPDATED: bars have different colors automatically ----
    bars = plt.bar(
        x_pos,
```

```
        test_means,
        yerr=test_stds,
        capsize=5,
        alpha=0.7
    )
    for i, b in enumerate(bars):
        b.set_color(f"C{i}")
    # ------------------------------------------------------------

    plt.xlabel("Performance Metrics")
    plt.ylabel("Score")
    plt.title(f"{len(metrics_names)} Metrics Cross-Validation Performance␣
↪({prefix.upper()})")
    plt.xticks(x_pos, [m.upper() for m in metrics_names], rotation=0)
    plt.ylim(0, 1)
    plt.grid(True, alpha=0.3, axis="y")

    for i, (mean, std) in enumerate(zip(test_means, test_stds)):
        plt.text(i, mean + 0.02, f"{mean:.3f}±{std:.3f}", ha="center",␣
↪va="bottom", fontsize=10)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_cross_validation_results.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()


def create_comprehensive_visualizations(y_test, y_proba, model, feature_names,␣
↪output_dir, prefix, cv_metrics=None):
    os.makedirs(output_dir, exist_ok=True)

    rf_model = model.named_steps["rf"]
    imp_df = pd.DataFrame({
        "feature": feature_names,
        "importance": rf_model.feature_importances_
    }).sort_values("importance", ascending=False)

    imp_df.to_csv(f"{output_dir}/{prefix}_feature_importances.csv", index=False)

    # Feature importance plot
    plt.figure(figsize=(12, 8))
    top_features = imp_df.head(15).iloc[::-1]
    plt.barh(range(len(top_features)), top_features["importance"], alpha=0.7)
    plt.yticks(range(len(top_features)), top_features["feature"])
    plt.xlabel("Feature Importance Score")
    plt.title(f"Top 15 Features for {prefix.upper()} Prediction")
    plt.grid(True, alpha=0.3, axis="x")
```

```
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_feature_importances.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # ROC + PR curves
    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

    fpr, tpr, _ = roc_curve(y_test, y_proba)
    roc_auc = roc_auc_score(y_test, y_proba)

    # ---- UPDATED: ROC curve line color ----
    ax1.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}", linewidth=2, color="C0")
    # ---------------------------------------

    ax1.plot([0, 1], [0, 1], "k--", alpha=0.5)
    ax1.set_xlabel("False Positive Rate")
    ax1.set_ylabel("True Positive Rate")
    ax1.set_title(f"ROC Curve ({prefix.upper()})")
    ax1.legend()
    ax1.grid(True, alpha=0.3)

    precision_vals, recall_vals, _ = precision_recall_curve(y_test, y_proba)
    pr_auc = average_precision_score(y_test, y_proba)

    # ---- UPDATED: PR curve line color ----
    ax2.plot(recall_vals, precision_vals, label=f"AP = {pr_auc:.3f}",␣
↪linewidth=2, color="C1")
    # --------------------------------------

    ax2.set_xlabel("Recall")
    ax2.set_ylabel("Precision")
    ax2.set_title(f"Precision-Recall Curve ({prefix.upper()})")
    ax2.legend()
    ax2.grid(True, alpha=0.3)

    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_roc_pr_curves.png", dpi=200,␣
↪bbox_inches="tight")
    plt.show()

    # Confusion matrix heatmap (threshold 0.5)
    y_pred = (y_proba >= 0.5).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
```

```
                xticklabels=["Pred 0", "Pred 1"],
                yticklabels=["True 0", "True 1"])
    plt.title(f"Confusion Matrix ({prefix.upper()})")
    plt.ylabel("True Label")
    plt.xlabel("Predicted Label")
    plt.tight_layout()
    plt.savefig(f"{output_dir}/{prefix}_confusion_matrix.png", dpi=200,␣
 ↪bbox_inches="tight")
    plt.show()

    if cv_metrics:
        create_validation_comparison_plot(cv_metrics, output_dir, prefix)

    return imp_df


# ============================================================
# 8) SAVE RESULTS (GENERALIZED)
# ============================================================
def save_comprehensive_results(model, metrics, imp_df, df_clean, target_col,␣
 ↪feature_names, output_dir, prefix, cv_metrics=None):
    print(f"\n SAVING OUTPUTS to {output_dir}/ ...")
    os.makedirs(output_dir, exist_ok=True)

    # Metrics text file
    with open(f"{output_dir}/{prefix}_comprehensive_metrics.txt", "w",␣
 ↪encoding="utf-8") as f:
        f.write(f"COMPREHENSIVE RANDOM FOREST RESULTS ({prefix.upper()})\n")
        f.write("=" * 60 + "\n")
        f.write(f"Dataset shape: {df_clean.shape}\n")
        f.write(f"Target column: {target_col}\n")
        f.write(f"Features used: {len(feature_names)}\n\n")

        f.write("TEST SET METRICS:\n")
        for k, v in metrics.items():
            f.write(f"{k}: {v:.4f}\n")
        f.write("\n")

        if cv_metrics:
            f.write("CROSS-VALIDATION (Mean ± Std):\n")
            for metric, scores in cv_metrics.items():
                f.write(f"{metric}: {scores['test_mean']:.4f} ±␣
 ↪{scores['test_std']:.4f}\n")
            f.write("\n")

        f.write("TOP 10 FEATURES:\n")
        for _, row in imp_df.head(10).iterrows():
```

```
            f.write(f"{row['feature']}: {row['importance']:.6f}\n")

    # Predictions on full dataset (IMPORTANT: use the same feature columns)
    X_full = df_clean.drop(columns=[target_col], errors="ignore")
    X_full = X_full.reindex(columns=feature_names)  # align to training feature␣
 ↪set

    preds = model.predict_proba(X_full)[:, 1]
    predictions_df = df_clean.copy()
    predictions_df[f"{prefix}_Pred_Prob"] = preds
    predictions_df[f"{prefix}_Pred"] = (preds >= 0.5).astype(int)
    predictions_df[f"{prefix}_Correct"] = (predictions_df[f"{prefix}_Pred"] ==␣
 ↪predictions_df[target_col]).astype(int)

    predictions_df.to_excel(f"{output_dir}/{prefix}_predictions.xlsx",␣
 ↪index=False)

    # Save model
    joblib.dump(model, f"{output_dir}/{prefix}_rf_model.joblib")

    print("  Saved metrics, plots, predictions, and model.")


# ============================================================
# 9) MAIN PIPELINE (ONE FUNCTION FOR ANY BFx)
# ============================================================
def run_comprehensive_bf_analysis(
    target_bf="BF1",
    use_cross_validation=True,
    use_validation_set=True,
    df=None
):
    """
    Run the full pipeline for a given target_bf (e.g., BF1, BF3, BF4, BF5).
    If df is None, it will prompt you to upload/load.
    """
    try:
        prefix = str(target_bf).strip().lower().replace(" ", "").replace("_",␣
 ↪"")
        output_dir = f"{prefix}_comprehensive_outputs"

        print("  COMPREHENSIVE MICROFINANCE ANALYSIS")
        print("=" * 60)
        print(f"  Target: {target_bf}")
        print(f"  Output folder: {output_dir}/")
        print("=" * 60)
```

```
        # Load data if df not provided
        if df is None:
            try:
                import google.colab  # noqa: F401
                df = load_excel_from_colab()
            except Exception:
                df = load_excel_locally()

        print(f"  Initial dataset shape: {df.shape}")

        # Detect + preprocess target
        target_col = detect_bf_target(df, target_bf=target_bf)
        y, valid_mask = preprocess_bf_target(df[target_col],␣
↪target_label=target_col)
        df_clean = df.loc[valid_mask].copy()
        print(f"  Clean dataset: {df_clean.shape} (after removing invalid␣
↪target rows)")

        # Features
        X_final, feature_names = build_feature_matrix(df_clean, target_col)

        # CV (10-fold by default) with 6 metrics
        cv_metrics = None
        if use_cross_validation:
            cv_metrics, _ = perform_cross_validation(X_final, y, cv_folds=10)

        # Train
        model, X_train, X_test, X_val, y_train, y_test, y_val =␣
↪train_with_comprehensive_validation(
            X_final, y, use_validation_set=use_validation_set
        )

        # Evaluate test
        _, y_proba, test_metrics = evaluate_model_comprehensive(model, X_test,␣
↪y_test, "Test")

        # (Optional) Evaluate val
        if X_val is not None and y_val is not None:
            evaluate_model_comprehensive(model, X_val, y_val, "Validation")

        # Visuals
        imp_df = create_comprehensive_visualizations(
            y_test=y_test,
            y_proba=y_proba,
            model=model,
            feature_names=feature_names,
            output_dir=output_dir,
```

```
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        # Save
        save_comprehensive_results(
            model=model,
            metrics=test_metrics,
            imp_df=imp_df,
            df_clean=df_clean,
            target_col=target_col,
            feature_names=feature_names,
            output_dir=output_dir,
            prefix=prefix,
            cv_metrics=cv_metrics
        )

        print("\n DONE.")
        return model, test_metrics, imp_df, cv_metrics

    except Exception as e:
        print(f"\n Analysis failed: {type(e).__name__}: {e}")
        import traceback
        traceback.print_exc(limit=2)
        return None, None, None, None


# ========= USAGE EXAMPLES =========
if __name__ == "__main__":
    # Single target run:
    run_comprehensive_bf_analysis(target_bf="BF6", use_cross_validation=True,␣
 ↪use_validation_set=True)

    # Multiple targets (run one by one):
    # for bf in ["BF1", "BF3", "BF4", "BF5"]:
    #     run_comprehensive_bf_analysis(target_bf=bf,␣
 ↪use_cross_validation=True, use_validation_set=True)
```

```
 COMPREHENSIVE MICROFINANCE ANALYSIS
============================================================
 Target: BF6
 Output folder: bf6_comprehensive_outputs/
============================================================
 Please select the Excel file with your encoded dataset…
```

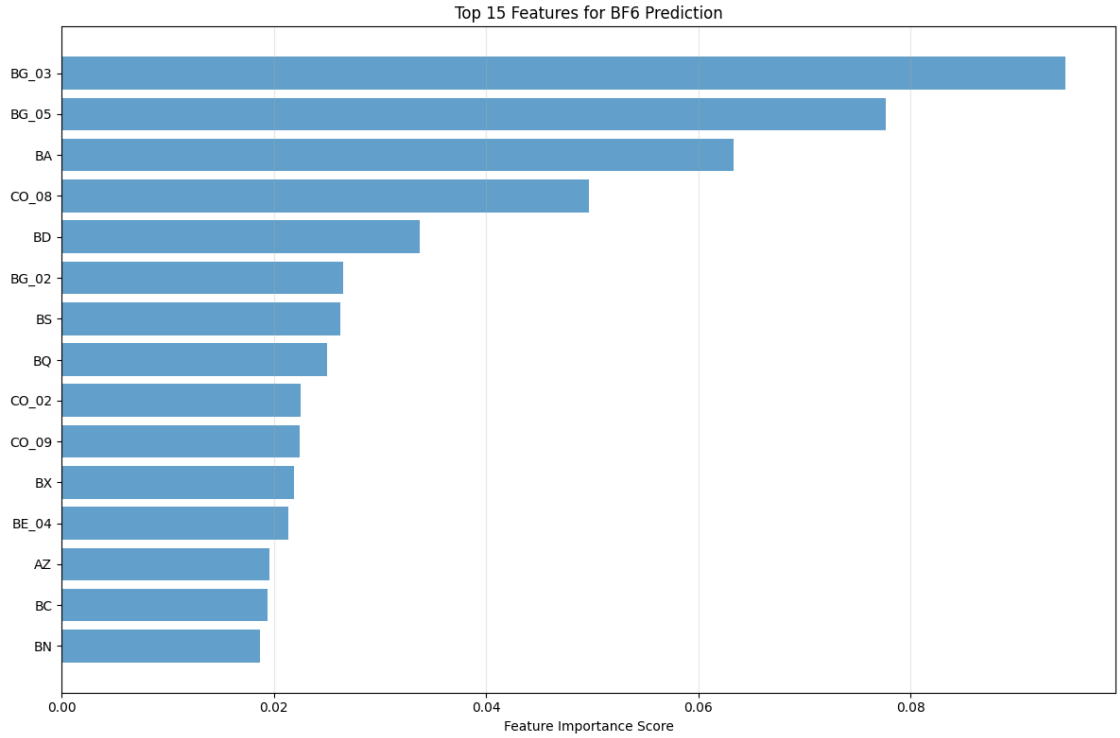
```
<IPython.core.display.HTML object>
```

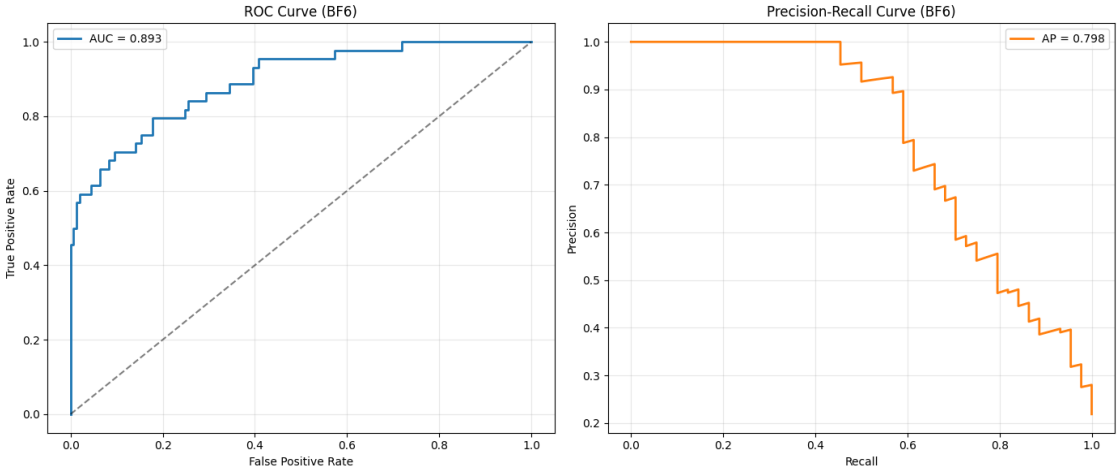
```
Saving rf_dataset_encoded_dataset.xlsx to rf_dataset_encoded_dataset (4).xlsx
```

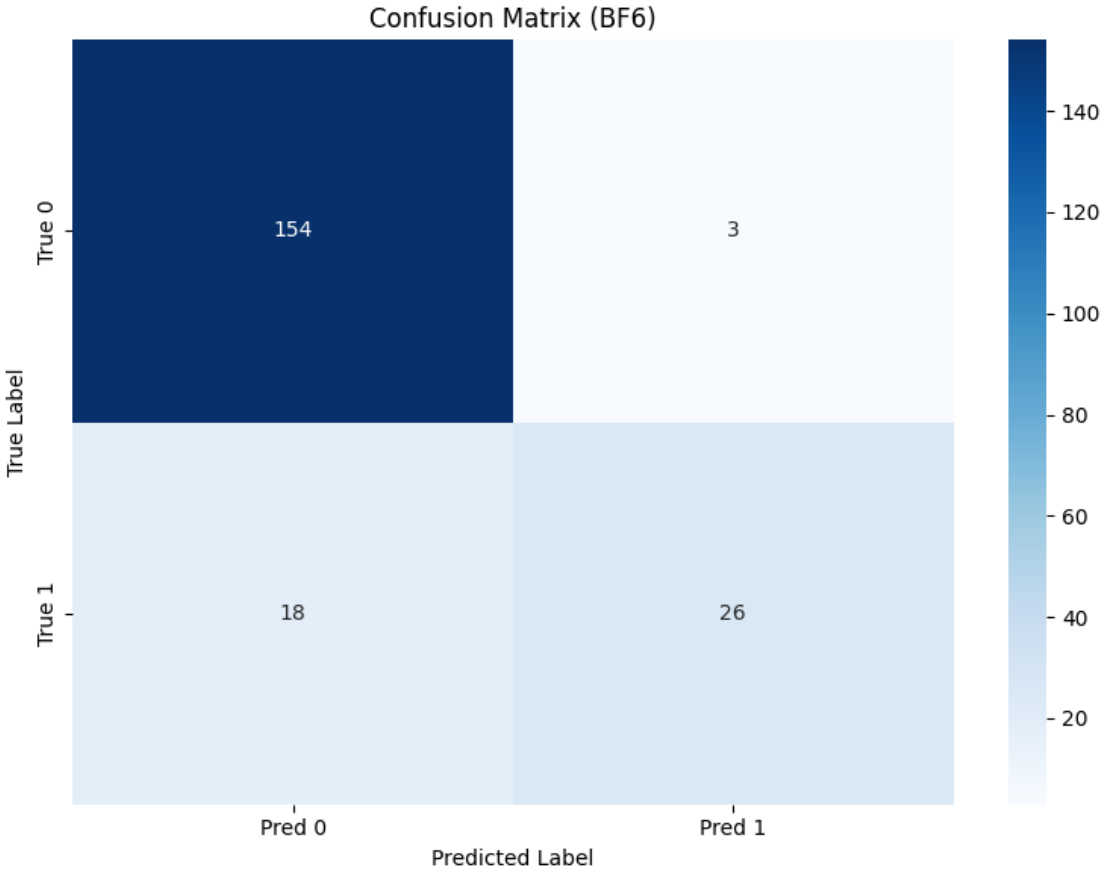
```
 Loaded dataset: (1002, 85) (rows, cols)
 Initial dataset shape: (1002, 85)
 BF6 column not found automatically.
 Available columns:
['D', 'AY', 'AZ', 'BA', 'BB', 'BC', 'BD', 'BE_01', 'BE_02', 'BE_03', 'BE_04',
'BE_05', 'BE_06', 'BE_07', 'BE_08', 'BE_09', 'BE_10', 'BF', 'BG_01', 'BG_02',
'BG_03', 'BG_04', 'BG_05', 'BG_06', 'BH', 'BI', 'BN', 'BO', 'BP', 'BQ', 'BR',
'BS', 'BT', 'BU', 'BV', 'BW_01', 'BW_02', 'BW_03', 'BW_04', 'BW_05', 'BX', 'BY',
'CB', 'CD', 'CE', 'CF', 'CG', 'CH', 'CI', 'CK', 'CL', 'CM_01', 'CM_02', 'CM_03',
'CM_04', 'CM_05', 'CM_06', 'CN', 'CO_01', 'CO_02', 'CO_03', 'CO_04', 'CO_05',
'CO_06', 'CO_07', 'CO_08', 'CO_09', 'CP', 'CR_01', 'CR_02', 'CR_03', 'CR_04',
'CS', 'CT_01', 'CT_02', 'CT_03', 'CT_04', 'CT_05', 'CT_06', 'CU', 'Occupation',
'Financial Behavior', 'Family', 'Lifestyle', 'Digital Literacy']
 Enter exact column name for BF6: BG_06
 Using specified target column: BG_06

 TARGET ANALYSIS: BG_06
----------------------------------------
Value distribution:
BG_06
0    783
1    219
Name: count, dtype: int64
Unique values: [np.int64(0), np.int64(1)]
 Final class balance: {0: 0.781, 1: 0.219}
 Clean dataset: (1002, 85) (after removing invalid target rows)

 FEATURE ENGINEERING
----------------------------------------
 Found 84 numeric features
 Handling missing values with median imputation…
 Final feature matrix: (1002, 84) (rows, cols)

 PERFORMING 10-FOLD CROSS-VALIDATION
==================================================
 CROSS-VALIDATION RESULTS (Mean ± Std):
----------------------------------------
ACCURACY       : 0.8862 ± 0.0348 (train: 0.9535)
PRECISION      : 0.7611 ± 0.0759 (train: 0.8832)
RECALL         : 0.6885 ± 0.1271 (train: 0.9077)
F1             : 0.7204 ± 0.1027 (train: 0.8951)
ROC_AUC        : 0.9140 ± 0.0417 (train: 0.9890)
AVERAGE_PRECISION: 0.8086 ± 0.0840 (train: 0.9582)

 Overfitting Analysis:
  Train-Test Accuracy Gap: 0.0673
    Potential overfitting detected
```

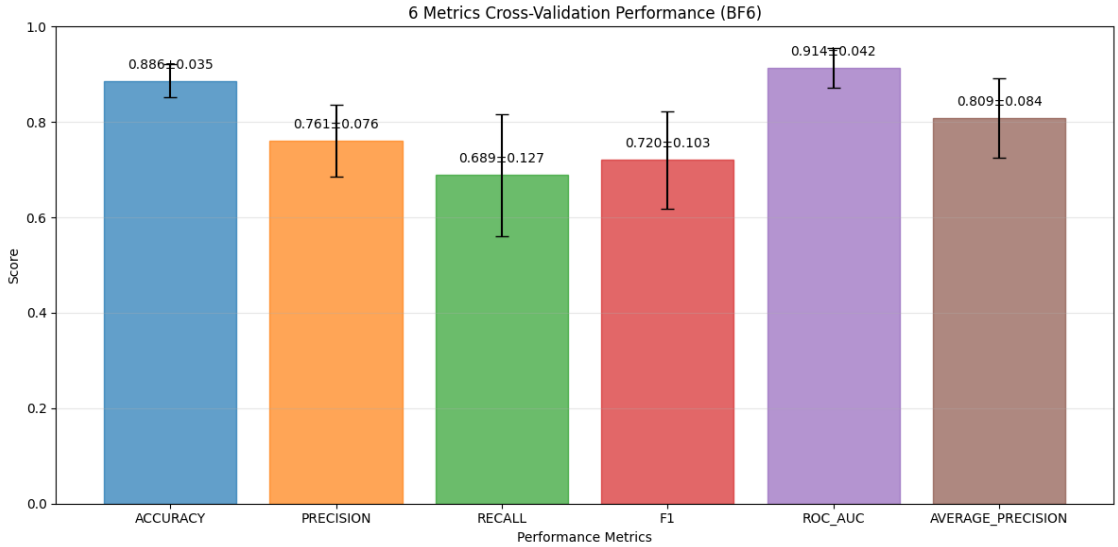
```
  MODEL TRAINING: Random Forest
--------------------------------------

  CREATING VALIDATION SET (20% of data)
==================================================
  Training set: 600 samples
  Test set: 201 samples
  Validation set: 201 samples
  Training model with regularization…

============================================================
  TEST SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.8955
  Precision: 0.8966
  Recall:    0.5909
  F1-Score:  0.7123
  ROC-AUC:   0.8932
  PR-AUC:    0.7982

  Test Classification Report:
              precision    recall  f1-score   support

           0      0.895     0.981     0.936       157
           1      0.897     0.591     0.712        44

    accuracy                          0.896       201
   macro avg      0.896     0.786     0.824       201
weighted avg      0.896     0.896     0.887       201

  Test Confusion Matrix:
[[TN FP]
 [FN TP]]
[[154   3]
 [ 18  26]]

============================================================
  VALIDATION SET COMPREHENSIVE EVALUATION
============================================================
  Accuracy:  0.8408
  Precision: 0.6667
  Recall:    0.5455
  F1-Score:  0.6000
  ROC-AUC:   0.8693
  PR-AUC:    0.6894

  Validation Classification Report:
              precision    recall  f1-score   support
```

```
              0       0.879     0.924     0.901       157
              1       0.667     0.545     0.600        44

    accuracy                            0.841       201
   macro avg        0.773     0.735     0.750       201
weighted avg        0.832     0.841     0.835       201

 Validation Confusion Matrix:
[[TN FP]
 [FN TP]]
[[145  12]
 [ 20  24]]
```




ROC Curve (BF6)
AUC = 0.893
True Positive Rate
False Positive Rate
Precision-Recall Curve (BF6)
AP = 0.798
Precision
Recall


Confusion Matrix (BF6)
154
3
18
26
True 0
True 1
True Label
Pred 0
Pred 1
Predicted Label




```
SAVING OUTPUTS to bf6_comprehensive_outputs/ …
Saved metrics, plots, predictions, and model.

DONE.
```

[1]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

[ ]:
```
# Install the package for Tex and then convert to PDF directly as LaTex
!sudo apt-get install texlive-xetex texlive-fonts-recommended␣
 ↪texlive-plain-generic pandoc
# Provide the file path of the notebook file inside the quotations
!jupyter nbconvert --to pdf "/content/drive/MyDrive/Colab Notebooks/CSE thesis␣
 ↪random forest P3.ipynb"
```

```
Reading package lists… Done
Building dependency tree… Done
Reading state information… Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
```

```
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries texlive-latex-base
  texlive-latex-extra texlive-latex-recommended texlive-pictures tipa
  xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
  poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
  fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
  fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
  | postscript-viewer perl-tk xpdf | pdf-viewer xzdec
  texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
  icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
  texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
  texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
  default-jre-headless tipa-doc
The following NEW packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-extra
  texlive-latex-recommended texlive-pictures texlive-plain-generic
  texlive-xetex tipa xfonts-encodings xfonts-utils
0 upgraded, 53 newly installed, 0 to remove and 2 not upgraded.
Need to get 182 MB of archives.
After this operation, 571 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
```

```
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.11 [50.1 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 http://archive.ubuntu.com/ubuntu jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsynctex2
amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
```

```
Get:33 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 6s (29.2 MB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based
frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 78,
<> line 53.)
debconf: falling back to frontend: Readline
```

```
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package fonts-droid-fallback.
(Reading database … 117540 files and directories currently installed.)
Preparing to unpack …/00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
…
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) …
Selecting previously unselected package fonts-lato.
Preparing to unpack …/01-fonts-lato_2.0-2.1_all.deb …
Unpacking fonts-lato (2.0-2.1) …
Selecting previously unselected package poppler-data.
Preparing to unpack …/02-poppler-data_0.4.11-1_all.deb …
Unpacking poppler-data (0.4.11-1) …
Selecting previously unselected package tex-common.
Preparing to unpack …/03-tex-common_6.17_all.deb …
Unpacking tex-common (6.17) …
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack …/04-fonts-urw-base35_20200910-1_all.deb …
Unpacking fonts-urw-base35 (20200910-1) …
Selecting previously unselected package libgs9-common.
Preparing to unpack …/05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb …
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libidn12:amd64.
Preparing to unpack …/06-libidn12_1.38-4ubuntu1_amd64.deb …
Unpacking libidn12:amd64 (1.38-4ubuntu1) …
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack …/07-libijs-0.35_0.35-15build2_amd64.deb …
Unpacking libijs-0.35:amd64 (0.35-15build2) …
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack …/08-libjbig2dec0_0.19-3build2_amd64.deb …
Unpacking libjbig2dec0:amd64 (0.19-3build2) …
Selecting previously unselected package libgs9:amd64.
Preparing to unpack …/09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb …
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) …
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack …/10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack …/11-libwoff1_1.0.2-1build4_amd64.deb …
Unpacking libwoff1:amd64 (1.0.2-1build4) …
Selecting previously unselected package dvisvgm.
Preparing to unpack …/12-dvisvgm_2.13.1-1_amd64.deb …
Unpacking dvisvgm (2.13.1-1) …
Selecting previously unselected package fonts-lmodern.
Preparing to unpack …/13-fonts-lmodern_2.004.5-6.1_all.deb …
```

```
Unpacking fonts-lmodern (2.004.5-6.1) …
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack …/14-fonts-noto-mono_20201225-1build1_all.deb …
Unpacking fonts-noto-mono (20201225-1build1) …
Selecting previously unselected package fonts-texgyre.
Preparing to unpack …/15-fonts-texgyre_20180621-3.1_all.deb …
Unpacking fonts-texgyre (20180621-3.1) …
Selecting previously unselected package libapache-pom-java.
Preparing to unpack …/16-libapache-pom-java_18-1_all.deb …
Unpacking libapache-pom-java (18-1) …
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack …/17-libcommons-parent-java_43-1_all.deb …
Unpacking libcommons-parent-java (43-1) …
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack …/18-libcommons-logging-java_1.2-2_all.deb …
Unpacking libcommons-logging-java (1.2-2) …
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack …/19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package rubygems-integration.
Preparing to unpack …/20-rubygems-integration_1.18_all.deb …
Unpacking rubygems-integration (1.18) …
Selecting previously unselected package ruby3.0.
Preparing to unpack …/21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb …
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) …
Selecting previously unselected package ruby-rubygems.
Preparing to unpack …/22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb …
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) …
Selecting previously unselected package ruby.
Preparing to unpack …/23-ruby_1%3a3.0~exp1_amd64.deb …
Unpacking ruby (1:3.0~exp1) …
Selecting previously unselected package rake.
Preparing to unpack …/24-rake_13.0.6-2_all.deb …
Unpacking rake (13.0.6-2) …
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack …/25-ruby-net-telnet_0.1.1-2_all.deb …
Unpacking ruby-net-telnet (0.1.1-2) …
Selecting previously unselected package ruby-webrick.
Preparing to unpack …/26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb …
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) …
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack …/27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb …
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) …
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack …/28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb …
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) …
Selecting previously unselected package libsynctex2:amd64.
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Preparing to unpack …/29-libsynctex2_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libsynctex2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack …/30-libteckit0_2.5.11+ds1-1_amd64.deb …
Unpacking libteckit0:amd64 (2.5.11+ds1-1) …
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack …/31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
…
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
…/32-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb …
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package libzzip-0-13:amd64.
Preparing to unpack …/33-libzzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb …
Unpacking libzzip-0-13:amd64 (0.13.72+dfsg.1-1.1) …
Selecting previously unselected package xfonts-encodings.
Preparing to unpack …/34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb …
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) …
Selecting previously unselected package xfonts-utils.
Preparing to unpack …/35-xfonts-utils_1%3a7.7+6build2_amd64.deb …
Unpacking xfonts-utils (1:7.7+6build2) …
Selecting previously unselected package lmodern.
Preparing to unpack …/36-lmodern_2.004.5-6.1_all.deb …
Unpacking lmodern (2.004.5-6.1) …
Selecting previously unselected package preview-latex-style.
Preparing to unpack …/37-preview-latex-style_12.2-1ubuntu1_all.deb …
Unpacking preview-latex-style (12.2-1ubuntu1) …
Selecting previously unselected package t1utils.
Preparing to unpack …/38-t1utils_1.41-4build2_amd64.deb …
Unpacking t1utils (1.41-4build2) …
Selecting previously unselected package teckit.
Preparing to unpack …/39-teckit_2.5.11+ds1-1_amd64.deb …
Unpacking teckit (2.5.11+ds1-1) …
Selecting previously unselected package tex-gyre.
Preparing to unpack …/40-tex-gyre_20180621-3.1_all.deb …
Unpacking tex-gyre (20180621-3.1) …
Selecting previously unselected package texlive-binaries.
Preparing to unpack …/41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb …
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) …
Selecting previously unselected package texlive-base.
Preparing to unpack …/42-texlive-base_2021.20220204-1_all.deb …
Unpacking texlive-base (2021.20220204-1) …
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack …/43-texlive-fonts-recommended_2021.20220204-1_all.deb …
Unpacking texlive-fonts-recommended (2021.20220204-1) …
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Selecting previously unselected package texlive-latex-base.
Preparing to unpack …/44-texlive-latex-base_2021.20220204-1_all.deb …
Unpacking texlive-latex-base (2021.20220204-1) …
Selecting previously unselected package libfontbox-java.
Preparing to unpack …/45-libfontbox-java_1%3a1.8.16-2_all.deb …
Unpacking libfontbox-java (1:1.8.16-2) …
Selecting previously unselected package libpdfbox-java.
Preparing to unpack …/46-libpdfbox-java_1%3a1.8.16-2_all.deb …
Unpacking libpdfbox-java (1:1.8.16-2) …
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack …/47-texlive-latex-recommended_2021.20220204-1_all.deb …
Unpacking texlive-latex-recommended (2021.20220204-1) …
Selecting previously unselected package texlive-pictures.
Preparing to unpack …/48-texlive-pictures_2021.20220204-1_all.deb …
Unpacking texlive-pictures (2021.20220204-1) …
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack …/49-texlive-latex-extra_2021.20220204-1_all.deb …
Unpacking texlive-latex-extra (2021.20220204-1) …
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack …/50-texlive-plain-generic_2021.20220204-1_all.deb …
Unpacking texlive-plain-generic (2021.20220204-1) …
Selecting previously unselected package tipa.
Preparing to unpack …/51-tipa_2%3a1.3-21_all.deb …
Unpacking tipa (2:1.3-21) …
Selecting previously unselected package texlive-xetex.
Preparing to unpack …/52-texlive-xetex_2021.20220204-1_all.deb …
Unpacking texlive-xetex (2021.20220204-1) …
```